Quantifying the Sensitivity of EU Debt Sustainability Analysis

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Abstract

The reformed EU fiscal rules derive each country's required fiscal adjustment from a debt sustainability analysis (DSA). Using a replication of the Commission's 2024 reference trajectories, this paper quantifies how the resulting fiscal targets depend on the inputs of the DSA. Because the target is the highest balance required by several criteria, an input matters only where it affects the binding criterion. I draw forecasts and parameters jointly from past forecast errors and published estimates. The target rises by 0.65 percentage points of GDP at the 75th percentile and falls by 0.36 points at the 25th. The average required adjustment is 2.6 points over four years. With crisis-level multipliers well above one, the debt sustainability safeguard can require implausibly large adjustments or become impossible to meet. The stochastic criterion rarely binds, so its specification matters little. Modeling choices reflecting the degree of torreble risk can move the targets about as much as the uncertainty about forecasts and parameters do.

JEL classification: E62, H62, H63, H68

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1 Introduction

Since April 2024, the EU fiscal rules have set fiscal targets for each member state through a debt sustainability analysis (DSA). Member states with government debt above 60 percent of GDP or a deficit above 3 percent of GDP receive from the European Commission a reference trajectory for net expenditure over an adjustment period of four to seven years. The trajectory leads to a target for the structural primary balance (SPB), the primary balance corrected for the cycle and one-off measures, at the end of the adjustment period. The Commission sets the SPB target so that debt declines, or stays below 60 percent of GDP, in a baseline projection, in stress tests and in a stochastic simulation. Numerical safeguards and the excessive deficit procedure (EDP) can raise the target further.

The DSA makes these targets country-specific, but it also makes them depend on multiple forecasts, parameters and methodological choices. Governments can argue for different inputs in their medium-term fiscal plans, at times with large consequences for the required adjustment (Zettelmeyer et al., 2025), and the regulation foresees a review of the DSA methodology (European Parliament and Council of the European Union, 2024; Boivin and Darvas, 2025). Both the assessment of national plans and the review require an understanding of how much the targets depend on each input.

This paper quantifies the sensitivity of the SPB targets of the 27 EU member states to the inputs of the Commission's DSA. It separates three questions. The first is how much the targets depend on economic forecasts and model parameters. The second is how much they depend on the method of the stochastic analysis. The third is how much they depend on the calibration of risk tolerance, that is, on how much risk the rules accept. It uses the replication of the DSA that we present in Darvas et al. (2023, 2024). This replication reproduces the reference trajectories of the Commission's prior guidance of 2024. I divide the inputs into four groups: forecasts, model parameters, the specification of the stochastic analysis, and the calibration of risk tolerance. I vary inputs both individually and jointly, by taking draws from the distributions of forecast errors and published parameters. For each change, I report the effect on the target required by the DSA criteria, as well as on the binding target, which in addition meets the safeguards and the minimum adjustment of the EDP. Because the target is the highest of several requirements, an input moves it only through the binding criterion. In the baseline, the deficit criterion and the adverse interest-growth stress test determine the DSA target for 23 of the 27 countries. The effects of the inputs are therefore concentrated on these two criteria. Because a favourable input lowers the target only until another criterion binds, whereas an adverse input raises it without such a limit, inputs affect targets asymmetrically.

The first group of model inputs contains the forecasts that determine the central projection of debt: interest rates and inflation ten years ahead, potential growth, and ageing costs. The second group contains the model parameters that link the SPB to output: the fiscal multiplier, the persistence of the multiplier effect, and the budget semi-elasticity. I first shift each forecast

by the size of a typical past forecast error and vary each parameter within the range of published estimates. Potential growth and the long-term interest rate ten years ahead have the largest effects. I then draw the forecasts and the parameters jointly from the distribution of past forecast errors and published parameter estimates. I define favourable and adverse combinations of assumptions as the 25th and 75th percentiles of the resulting targets. For the 21 countries with positive adjustment requirements, the favourable combination lowers the DSA target by 0.36 percentage points of GDP on average and the adverse combination raises it by 0.65 points. The average required baseline adjustment is 2.6 points over four years. Per year, the two changes amount to about 0.09 and 0.16 points. The adverse change over four years is thus about as large as the cumulative threshold for deviations in the control account (Council of the European Union, 2024). Because I remove the average error from each series, the draws measure the spread of past forecast errors but not a persistent bias in the forecasts.

The stochastic criterion, which featured prominently in previous reform proposals (Blanchard et al., 2021), only sets the target for Finland. As a result, planned revisions of the stochastic DSA by the Commission, which would allow for persistent and non-normal shocks (Bec et al., 2025), affect average targets little. A VAR bootstrap that addresses both points raises the DSA target of the countries with adjustment requirements by 0.02 points on average. It changes the target substantially only for Finland.

The calibration of assumptions that determine how much risk policymakers are willing to tolerate has considerable effects on targets. These assumptions are the probability threshold of the stochastic criterion, the size of the stress tests and the period for covering rising ageing costs. I refer to these settings as the risk tolerance. Within the alternatives I consider, these settings can move the targets by amounts comparable to the uncertainty about forecasts and parameters. A strict combination of assumption increases target sby 0.18, while a lenient lowers it by 0.4. For the 21 countries with adjustment requirements, adverse interest-growth shocks of 0.75 instead of 0.5 percentage points raise the DSA target by 0.16 points. Comparing the shock size in the adverse $r - g$ test with past forecast errors shows that the baseline shock size of 0.5 percentage point is exceeded in 30 percent of cases. The test thus roughly corresponds to insuring against the 70 percent of cases in which forecasts overestimate $r - g$. Although this is the same level as the stochastic probability threshold, the $r - g$ stress test target exceeds the stochastic target in most cases.

Assumptions about the fiscal multiplier matter mostly through their interaction with the debt sustainability safeguard. The safeguard requires the debt ratio to decline by a minimum amount during the adjustment. A larger multiplier slows this decline, so that the binding targets of Finland, Austria and Germany rise with the multiplier. With an exceptionally high multiplier of 1.5, typically estimated for crisis conditions, Finland and Spain cannot meet the safeguard at all. The multiplier mainly delays the decline in debt, however. After the adjustment, debt falls at a similar pace whatever the multiplier.

A repayment profile based on Eurostat's residual debt maturity data primarily affects the

targets of Greece and Finland. Repeating the analysis with input data from September 2026 barely changes the average effect of each input, although the binding criterion changes for several countries.

The results have three implications for the review of the DSA. First, the size of the stress tests and the ageing cost period deserve as much attention as the stochastic simulations. Second, the debt sustainability safeguard's test of debt decline during the adjustment can introduce adverse feedback via the fiscal multiplier. Third, the level of risk that the rules implicitly accept via stress tests and probability thresholds should be justified explicitly.

The paper relates to a growing literature that evaluates the reformed EU rules and the role of the DSA in them (Heimberger, 2023; Pench, 2024, 2025; Cottarelli, 2024; Wieland, 2024; Jousten, 2025; Erce, 2025; Gros and Hofer, 2026; Boivin and Darvas, 2025; European Fiscal Board, 2025). In Darvas et al. (2023, 2024), we use a replication of the Commission's DSA to evaluate the Commission's proposal and the adopted rules. Several of these studies identify the inputs of the DSA as a source of uncertainty or discretion, but most do not quantify how much the targets depend on them. Other work, including our own, examines single inputs: the size and persistence of the fiscal multiplier (Heimberger et al., 2024), demographic risks (Darvas et al., 2025b), the growth assumptions in Germany's medium-term plan (Zettelmeyer et al., 2025), the specification of the stochastic DSA (Darvas et al., 2025a), and the effect of public investment and structural reforms on potential growth (Gottschalk et al., 2025). This paper complements these studies by quantifying the effects of a wider set of inputs in one framework, with ranges based on past forecast errors and published estimates.

The rest of the paper is organised as follows. Section 2 describes how the DSA sets the fiscal adjustment targets and classifies the inputs. Section 3 reports the effects of the forecasts and the model parameters, including the joint draws. Section 4 turns to the stochastic analysis, the stress tests and the calibration of risk tolerance, and compares their effects with those of the forecasts and parameters. Section 5 covers the repayment profile of debt and repeats the checks with newer data. Section 6 concludes and discusses channels that the DSA leaves out.

2 Fiscal adjustment targets

2.1 Debt sustainability analysis in the EU fiscal rules

The baseline of this sensitivity analysis is the replication of the Commission's DSA that we present in Darvas et al. (2024). The DSA model projects the debt ratio with the standard debt-dynamics identity (European Commission, 2024b). The central model variables are the nominal growth, the primary balance, and the average interest rate on debt. They depend on the SPB path through a fiscal multiplier and the budget semi-elasticity (Carnot and de Castro, 2015; Mourre

et al., 2019).[1] Annex 2 of Darvas et al. (2024) and the repository describe the method in detail. The input data used for the baseline projection are taken from the 2024 Commission's prior guidance calculation sheets. The reference year is therefore 2024 and the four-year adjustment spans the period from 2025 to 2028.

DSA determines the target level for the SPB. This SPB target is the lowest SPB at the end of the adjustment period that meets a given set of criteria. For the purpose of the sensitivity anlysis, I differentiate between the *DSA target*, the SPB that meets the criteria of the DSA alone, and the *binding target*, which in addition is compliant with safeguards and the EDP. There are three kinds of DSA criteria. Under the deterministic debt criteria, the debt ratio must decline, or stay below 60 percent of GDP, over the ten years after the adjustment period. This must hold in a baseline scenario and in three stress tests. Debt declines if it falls in each of these ten years. The stress tests assume a lower SPB, a higher interest-growth differential $(r-g)$ and a temporary rise in market interest rates. Under the deficit criterion, the deficit must stay below 3 percent of GDP over the same ten years. Under the stochastic criterion, the debt ratio must decline over the five years after the adjustment period with a probability of at least 70 percent. The model computes this probability from simulated shocks to growth, interest rates, exchange rates, and the primary balance.

The binding target must meet four constraints in addition to the DSA criteria. These are an adjustment floor described below, the minimum adjustment under the EDP, the debt sustainability safeguard, and the deficit resilience safeguard. The debt sustainability safeguard sets a minimum average decline in debt over the adjustment years outside the EDP. Debt must fall by at least 1 percentage point of GDP per year while it exceeds 90 percent of GDP. It must fall by at least 0.5 points per year while it lies between 60 and 90 percent. The deficit resilience safeguard requires a minimum annual adjustment until the structural deficit is below 1.5 percent of GDP.

I implement these rules in the same way as in the Commission's prior guidance calculation sheets. Several features that are particular to this implementation should be noted. First, the debt criteria apply in one of two variants to all scenarios at once. In the first variant, debt declines in the baseline and in the three stress tests. It also declines with a probability of at least 70 percent in the stochastic analysis. In the second variant, debt stays below 60 percent of GDP in the baseline and in the three stress tests. The less demanding variant applies. Second, the annual adjustment is constant across the adjustment period, although the rules would permit an adjustment that is larger in the early years. Together with the safeguards, such front-loaded adjustment can lower the total adjustment in some cases. [2] Third, the annual adjustment features what I call an adjustment floor, i.e., adjustment cannot be negative for countries that receive a reference trajectory. Fourth, countries with debt below 60 percent and a deficit below 3 percent of GDP, receive only technical information. For these countries, only the deficit criterion and the

[1] The budget semi-elasticity is the change in the budget balance, in percent of GDP, when output changes by 1 percent.

[2] See the accompanying GitHub repository for a comparrison of implemenations with and without front-loading.

Table 1: Baseline SPB targets and Commission prior guidance, four-year adjustment (percent of GDP)

	2024		DSA target		Binding target			
	Debt	SPB	Comm.	Model	Comm.	Model	DSA criterion	Binding constraint
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Greece	153.9	1.73	2.37	2.33	2.61	2.58	Adverse $r-g$ test: debt declines	Deficit resilience
Italy	138.6	−1.05	3.35	3.31	3.35	3.31	Adverse $r-g$ test: debt declines	DSA criteria
France	112.4	−2.97	0.79	0.75	0.79	0.75	Adverse $r-g$ test: debt declines	DSA criteria
Spain	105.5	−0.85	2.59	2.59	2.59	2.59	Adverse $r-g$ test: debt declines	DSA criteria
Belgium	103.4	−2.03	1.25	1.21	1.25	1.21	Adverse $r-g$ test: debt declines	DSA criteria
Portugal	95.6	2.20	2.52	2.52	2.52	2.52	Adverse $r-g$ test: debt declines	DSA criteria
Finland	80.5	−0.53	2.11	1.83	5.87	4.83	Stochastic	Debt safeguard
Austria	79.5	−1.34	0.90	0.82	2.58	2.22	Adverse $r-g$ test: debt declines	Debt safeguard
Hungary	74.3	0.03	2.67	2.67	2.67	2.67	Adverse $r-g$ test: debt declines	DSA criteria
Cyprus	70.6	3.48	3.48	3.48	3.48	3.48	Adjustment floor	Adjustment floor
Slovenia	68.1	−1.24	0.52	0.52	0.52	0.52	Lower-SPB test: debt below 60% at end	DSA criteria
Germany	62.5	−1.02	1.14	1.02	2.74	2.46	Adverse $r-g$ test: debt below 60% at end	Debt safeguard
Croatia	59.5	−1.96	–	−0.44	−0.36	−0.36	Baseline: debt below 60% at end	Deficit resilience
Slovakia	58.5	−4.26	0.94	0.82	0.94	0.82	Lower-SPB test: debt below 60% at end	DSA criteria
Poland	53.7	−2.61	0.67	0.59	0.67	0.59	Deficit below 3%	DSA criteria
Malta	52.0	−2.90	−1.14	−1.22	−0.87	−0.87	Deficit below 3%	EDP minimum adjustment
Romania	50.9	−4.41	0.47	0.27	0.47	0.27	Deficit below 3%	DSA criteria
Netherlands	47.1	−0.64	–	0.08	0.08	0.08	Deficit below 3%	DSA criteria
Czechia	45.2	−0.09	–	0.39	0.39	0.39	Deficit below 3%	DSA criteria
Latvia	44.5	−1.36	–	−1.24	−0.14	−0.13	Deficit below 3%	Deficit resilience
Ireland	42.5	2.53	–	−0.63	–	−0.63	Deficit below 3%	DSA criteria
Lithuania	38.3	−0.37	–	0.27	0.27	0.27	Deficit below 3%	DSA criteria
Sweden	32.0	0.69	–	−1.75	−0.83	−0.74	Deficit below 3%	Deficit resilience
Luxembourg	27.1	0.10	–	−0.22	–	−0.22	Deficit below 3%	DSA criteria
Denmark	26.5	2.91	–	−1.09	−1.09	−1.09	Adjustment floor	Adjustment floor
Bulgaria	24.5	−2.09	–	−2.05	−0.87	−0.88	Deficit below 3%	Deficit resilience
Estonia	21.4	−0.29	−0.29	−0.29	−0.29	−0.29	Adjustment floor	Adjustment floor

Source: Author's calculations based on the European Commission's prior guidance calculation sheets of 2024.
Note: SPB at the end of the adjustment period (2028). Columns (3) and (5) are the Commission's targets. Column (3) is the SPB in 2024 plus four times the DSA-based annual adjustment. Countries receiving technical information report no DSA-based adjustment. Columns (4) and (6) are the model baseline. Columns (3), (4) and (7) follow the Commission's definition of the DSA target, which includes the adjustment floor and the rounding of the annual adjustment; in the rest of the paper, the DSA target excludes both. Column (7) is the DSA criterion that sets the DSA target. Column (8) is the constraint that sets the binding target: the DSA criteria, the adjustment floor, the debt sustainability safeguard, the deficit resilience safeguard or the minimum adjustment of the EDP. Countries are ordered by debt in 2024.

requirement that debt stays below 60 percent in the baseline scenario apply and their adjustment floor is -1 percentage point per year.

An additional feature particular to the Comission's implementation concerns the fiscal mutlipler and the output gap clouse. During the adjustment period, the Commission calcualtion sheets describe the output gap following a fiscal adjustment as the sum of the non-adjustment baseline output gap and two-thirds of the difference between the previous year's output gap and its baseline. After the adjustment period, the gap closes linearly over three years. I instead use a more intutive version of the output gap closure, in which the deviation in the output gap form its baseline in any given year is the sum of the fiscal mutliplier effects, i.e., the change in SPB multiplied by the fiscal multipler, of previous years. For a fiscal multiplier persistence

of three years, the change in the output gap in a given year t then is the sum of current years fiscal mutliplier effect, two-thirds of the effect in $t-1$, and one-third of the effect in $t-2$. This implementation makes a change to the persistence straight forward. For a 4-year persistence, the change in the output gap would be the sum of the full effect in t up to one-fourth of the fiscal multiplier effect in $t-4$, for five years up to one-fifth in $t-5$, and so on. This implementation does not qualitatively affect the results of the sensitivty analysis. Table A6 in the Annex repeats the multiplier checks with the Commission's rule.

2.2 Replication of the Commission targets

Table 1 compares the baseline with the Commission's targets. The median absolute difference from the Commission's DSA targets is 0.04 percentage points.The largest difference is 0.28 points, for Finland. For the binding target, the median difference is 0.01 points.[3] The largest is 1.04 points, again for Finland, where the debt sustainability safeguard binds. The differene in the specification of the fiscal multiplier and output gap closure explained above, accounts for the gaps for the three countries in which the safeguard binds. For Finland, Austria and Germany, the model gives binding targets of 4.83, 2.22 and 2.46 percent of GDP. The Commission's sheets give 5.87, 2.58 and 2.74. With the Commission's rule for the output gap, the model gives 5.99, 2.58 and 2.78 (Table A6). These values are within 0.12 points of the Commission's targets. Section 3.3 shows that the multiplier results do not depend on this specification.

2.3 Determining the binding criterion

Each criterion requires its own SPB. The DSA target is the highest of these. Figure 1 shows the SPB that each DSA criterion and each constraint of the binding target requires in the baseline. In the baseline, the deficit criterion sets the DSA target for 14 countries. The adverse $r-g$ stress test sets it for nine countries and the lower-SPB stress test for two. The baseline debt criterion sets it for one country, and the stochastic criterion for Finland. For Cyprus, Denmark and Estonia, all DSA criteria require a lower SPB than in 2024. The adjustment floor therefore sets their binding target. These counts refer to the DSA target without the floor. Table 1 uses the Commission's definition, under which the floor sets the DSA target of these three countries. On that definition, the deficit criterion sets the DSA target for 11 countries.

Figure 1 shows the debt criteria in the variant that applies to each country. For Cyprus, Slovenia, Germany, Slovakia, Poland, Malta and Romania, the second variant is less demanding and therefore applies. In this variant, debt stays below 60 percent of GDP over the ten years after the adjustment period. Countries without a reference trajectory face only the deficit criterion and the requirement that debt stays below 60 percent in the baseline scenario.

[3]For this comparison, the minimum adjustment floor is part of both the DSA target and the binding target. All later analyses only feature this floor as part of the binding target.

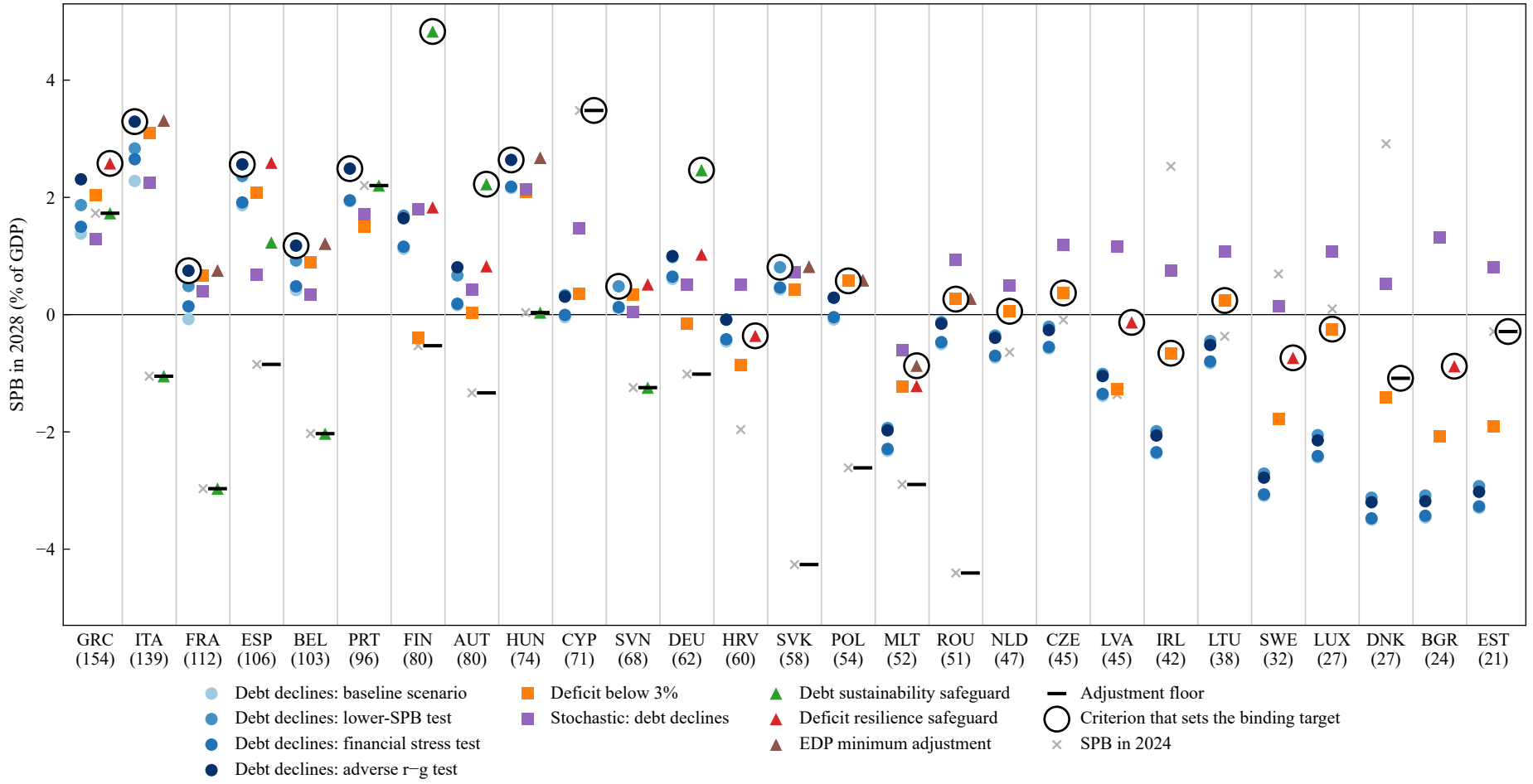


Figure 1: SPB required by each criterion in the baseline, four-year adjustment (percent of GDP)

Source: Author's calculations. Note: SPB at the end of the adjustment period (2028). Each column shows one country, ordered by debt in 2024 (in brackets), with the debt criteria (circles), the deficit and stochastic criteria (squares) and the constraints of the binding target (triangles) side by side. The open circle marks the criterion that sets the binding target. The bar shows the adjustment floor, for countries with a reference trajectory or where it binds. Grey crosses show the SPB in 2024. The debt criteria are shown in the variant that applies to the country. Section 2 defines the constraints.

For each constraint of the binding target, the figure shows the SPB that the constraint requires on its own. The figure shows the safeguard for countries with a reference trajectory and debt above 60 percent of GDP. For the deficit resilience safeguard and the minimum adjustment of the EDP, the figure adds their minimum steps to the path to the DSA target and shows associated targets only for the countries to which they apply. The adjustment floor is the SPB in 2024 for countries with a reference trajectory. For other countries, it is 4 percentage points below the SPB in 2024.

In the baseline, a constraint other than the DSA criteria determines the binding target in 12 countries. The debt sustainability safeguard determines it in Finland, Austria and Germany. The deficit resilience safeguard determines it in Greece, Croatia, Latvia, Sweden and Bulgaria, and the minimum adjustment of the EDP in Malta. The adjustment floor determines it in Cyprus, Denmark and Estonia.

An input changes the DSA target only if it moves the highest criterion or lifts another criterion above it. Inputs that act on a lower criterion leave the target unchanged. A favourable input lowers the target only until the next criterion binds. An adverse input, on the other hand, raises the target without such a limit.

To allow an assessemnt of the importance of assumptions both in isolation and in the context of other criteria, I report three distinct effects in the sensitivty analysis that follows. The first is the effect of each input on its *own criterion*, i.e., the criterion on which it acts directly. For the size of a stress test, this is the stress test itself. For the settings of the stochastic analysis, it is the stochastic criterion. For all other inputs, I use the debt criterion of the baseline scenario. The first level is the change in the SPB required by the own criterion. The second is the change in the DSA target, and the third is the change in the binding target.

2.4 Classification of the inputs

Table 2 lists inputs for the DSA in four groups. For each input, it shows where the input enters the projection, its baseline value, and the alternative values I consider. The groups separate three kinds of uncertainty. The first is uncertainty about the central projection itself. The second is uncertainty about future shocks around a given central projection. The third is how much of both kinds of uncertainty the rules accept.

The central projection is the projection of the baseline scenario. It rests on forecasts of a few key values, such as market interest rates, inflation and potential growth ten years ahead. The central projection also rests on model parameters, such as the fiscal multiplier. If the estimates or forecasts used for the central projections are off, so is the projected path of debt. Because such errors shift the baseline debt ratio projection, they affect every criterion and safeguard. The first two groups of DSA inputs capture this uncertainty about the central debt projection. To assess the effect of these inputs on SPB targets, I vary the forecasts by the size of past forecast errors and the parameters within the range of published estimates.

The stochastic analysis covers the second kind of uncertainty. The Commission DSA draws shocks to growth, interest rates, exchange rates and the primary balance from their historical distribution to simulates many debt paths around the central projection. The distribution of the simulated paths is called the fan chart. The shocks have a mean of zero, so the fan chart is centred on the central projection. In the Commission's current method, the shocks are also temporary. The fan chart thus describes deviations from year to year around a correct central projection. It does not describe a persistent error in the central projection, such as an overestimate of potential growth over ten years. The third group of DSA inputs assess the sensitivty of SPB targets to these assumptions. The methodological choices change the width and the shape of the fan chart but leave the central projection, and thus all deterministic criteria, unchanged. They affect the targets only through the stochastic criterion, which sets the target for one country in the baseline.

The fourth group of DSA inputs determines how much of the uncertainty described above is accepted by the rules. They include the probability threshold of the stochastic criterion, the size of the stress tests, and the period over which changes in ageing costs enter the projection. The probability threshold determines how much of the fan chart the target must cover. The stress tests shift the central projection by fixed amounts. The ageing cost period determines how far the rules rely on governments to offset changes in ageing costs elsewhere in the budget. The table also lists one data input, the repayment profile of debt, which Section 5 covers.

3 Forecasts and model parameters

3.1 Ranges of forecasts and model parameters

In the individual checks, I shift each forecast by about half of the interquartile range of past errors. The shifts are 1 percentage point for interest rates and potential growth and 0.75 points for inflation. For ageing costs, the shift is 0.2 points of GDP. Symmetric shifts show directly whether a target responds more to adverse than to favourable errors. In the joint draws of Section 3.4, I instead draw the errors from their full distribution. A forecast error is the realised value minus the forecast. The distribution is the same for all countries. Table A1 in the Annex reports the sources and percentiles of the errors.

For growth and inflation, I use the forecasts of autumn vintages of the IMF's World Economic Outlook (WEO) from 2000 to 2018 (Ech-charfi, 2024). For each vintage and EU country, I average the errors over the five forecast years. I then remove each country's average error. Before this step, growth was over-forecast by 0.7 percentage points per year on average. This average error largely reflects the decline in trend growth over the sample. This paper is concerned with the size of forecast errors. Official forecasts in the euro area have also been optimistic on average (Frankel and Schreger, 2013). Because I remove the average errors, the ranges do not cover the risk of a persistent bias in the central projection. The interquartile range is −0.84 to +1.08 percentage points for growth and −0.74 to +0.64 points for inflation. These are errors in actual

Table 2: Inputs of the DSA by group

Input	Where it enters	Baseline	Values checked
1. Forecasts: distribution of past forecast errors			
Real short-term rate, $T + 10$	Interest on new and refinanced debt	Commission	±1 pp.
Real long-term rate, $T + 10$	Interest on new and refinanced debt	Commission	±1 pp.
Inflation, $T + 10$	Nominal growth and rates	Commission	±0.75 pp.
Potential growth, $T + 10$	Real growth	Commission	±1 pp.
Ageing costs, change 2024–2038	SPB after the adjustment	Ageing Report	±0.2 pp. of GDP
Rates and inflation, $T + 30$	Long-run rates and growth	Commission	±0.5 and ±0.25 pp.[a]
2. Model parameters: range of published estimates			
Fiscal multiplier	Output effect of the adjustment	0.75	0.5 to 1.0; 1.5[b]
Persistence of the multiplier	Output effect of the adjustment	3 years	2 to 5 years
Budget semi-elasticity	Cyclical balance	Commission	×0.9 and ×1.1
3. Estimation of risk: alternative choices			
Start of stochastic projection	Stochastic criterion	After adjustment	First adjustment year
Horizon of stochastic projection	Stochastic criterion	5 years	10 years
Start of shock sample	Stochastic criterion	2000	1990, 2010
Shock frequency	Stochastic criterion	Quarterly	Annual
Shock distribution	Stochastic criterion	Normal	VAR bootstrap
Winsorisation of shocks	Stochastic criterion	5th/95th pct.	1st/99th pct., none
4. Risk tolerance: alternative settings			
Probability threshold	Stochastic criterion	70%	60%, 80%
Adverse $r - g$ shock	Stress test	0.5 pp.	0.25, 0.75 pp.
Financial stress shock	Stress test	1 pp.	0.5, 1.5 pp.
Lower-SPB shock	Stress test	0.5 pp.	0.25, 0.75 pp.
Ageing cost period	SPB after the adjustment	10 years	5 years
Data			
Repayment profile of debt	Refinancing	Historical average	Eurostat residual maturity

Note: Values of the forecasts are deviations from the baseline, in percentage points (pp.), of about the size of typical past errors; the joint draws use the full distribution of past errors. Forecasts and model parameters are varied one at a time and drawn jointly. All other inputs are varied one at a time. Table A1 in the Annex reports the sources of the forecast errors. [a] Interest rates ±0.5 pp., inflation ±0.25 pp.; individual checks only, since there is no evidence on errors at this horizon. [b] Individual check only, for crisis conditions.

rather than potential growth, and they include recessions. The range is therefore wide for a shift in potential growth.

For interest rates, the DSA takes market rates ten years ahead from forward rates. Only Germany has a public yield curve of that length. I therefore use the Bundesbank's zero-coupon curve for Germany. At each year end from 1986 to 2015, I compare its forward rates with the rates realised ten years later. Forward rates exceeded the realised 10-year rate by 3.0 percentage points on average. This bias reflects the term premium and the decline in rates since the 1990s, and I remove it. The interquartile range is −1.24 to +0.89 points for the long-term rate and −1.22

to +0.40 points for the short-term rate. The German curve leaves out errors in the spreads of other countries over Germany. These errors are largest for high-debt countries such as Italy and Greece. The ranges therefore likely understate interest rate risk for countries with high spreads. I also apply errors in nominal rates to real rates and vary inflation separately. The implied range of nominal rates is therefore somewhat wider than in the data.

For ageing costs, I use the eight scenarios of the 2024 Ageing Report that vary demographic, labour market and health cost assumptions (European Commission, 2024a). For each scenario, I compute the deviation from the baseline change in ageing costs between 2024 and 2038. The year 2038 is the end of the DSA horizon for a four-year adjustment. The interquartile range of these deviations is −0.24 to +0.14 percentage points of GDP.

T denotes the base year of the projection, which is 2024 in the baseline. I shift interest rates, inflation and potential growth at the anchor in $T + 10$, ten years after the base year. The shift rises linearly from zero in the last forecast year to its full size in $T + 10$. It then returns to zero by $T + 30$, in line with the Commission's interpolation between anchors. Real growth moves with potential growth, so that the output gap is unchanged. The shift in ageing costs grows linearly from 2024 and reaches its full size in 2038. The rate assumptions are real rates. When I shift inflation, nominal interest rates therefore move one for one with it.

The ranges of the model parameters come from the literature. The baseline multiplier is 0.75, as in Carnot and de Castro (2015). Most estimates of spending multipliers lie between 0.6 and 1 (Ramey, 2019; Ramey and Zubairy, 2018). Barro and Redlick (2011) find 0.4 to 0.5 on impact. I therefore vary the multiplier between 0.5 and 1.0, in the individual checks and in the joint draws. Higher multipliers apply under particular conditions. Multipliers were about 1.5 in the European consolidations after 2010 (Blanchard and Leigh, 2013). At that time, interest rates were at their lower bound and many countries consolidated together. Estimates for regions of a monetary union, where monetary policy does not respond, are also 1.5 or more (Nakamura and Steinsson, 2014; Chodorow-Reich, 2019). These estimates compare regions and capture spending financed from outside the region. They therefore do not directly measure the effect of a consolidation that a country finances through its own budget. I treat a multiplier of 1.5 as an additional individual check for such conditions. I do not include it in the joint draws. In the baseline, the effect of the multiplier fades over three years. I vary this persistence from two to five years. Cumulative multipliers over two years are at least as large as impact multipliers (Barro and Redlick, 2011; Ramey and Zubairy, 2018). Consolidations can also have lasting effects on output (Fatás and Summers, 2018; Gechert et al., 2019). Finally, I scale the budget semi-elasticity by 0.9 and 1.1. The largest revisions of the Commission's estimates between the 2014 and 2018 updates were about 10 percent (Mourre et al., 2019).

Table 3a: Change of the SPB relative to the baseline: forecasts and model parameters, four-year adjustment (percentage points of GDP)

		Own criterion		DSA target		Binding target	
Input	Value	Mean	Largest	Mean	Largest	Mean	Largest
1. Forecasts							
Real short-term rate, $T + 10$	−1 pp.	−0.04	−0.15 ITA	−0.04	−0.14 ITA	−0.03	−0.12 PRT
	+1 pp.	+0.04	+0.15 ITA	+0.04	+0.14 ITA	+0.03	+0.16 ITA
Real long-term rate, $T + 10$	−1 pp.	−0.26	−0.52 GRC	−0.23	−0.52 GRC	−0.17	−0.44 ESP
	+1 pp.	+0.25	+0.53 GRC	+0.23	+0.47 GRC	+0.17	+0.40 ITA
Inflation, $T + 10$	−0.75 pp.	+0.18	+0.46 ITA	+0.11	+0.45 ITA	+0.10	+0.44 ITA
	+0.75 pp.	−0.16	−0.45 ITA	−0.04	−0.26 FIN	−0.03	−0.20 HUN
Potential growth, $T + 10$	−1 pp.	+0.54	+1.23 ITA	+0.42	+1.21 ITA	+0.35	+1.20 ITA
	+1 pp.	−0.51	−1.18 ITA	−0.29	−0.65 ESP	−0.20	−0.64 ESP
Ageing costs	−0.2 pp.	−0.09	−0.15 AUT	−0.08	−0.14 AUT	−0.06	−0.16 BEL
	+0.2 pp.	+0.09	+0.15 AUT	+0.08	+0.14 AUT	+0.06	+0.12 ESP
Real short-term rate, $T + 30$	−0.5 pp.	0.00	−0.01 PRT	0.00	−0.01 PRT	0.00	−0.04 PRT
	+0.5 pp.	0.00	+0.01 PRT	0.00	+0.01 PRT	0.00	0.00
Real long-term rate, $T + 30$	−0.5 pp.	−0.01	−0.03 GRC	−0.01	−0.02 GRC	−0.01	−0.04 PRT
	+0.5 pp.	+0.01	+0.03 GRC	+0.01	+0.02 GRC	0.00	+0.03 GRC
Inflation, $T + 30$	−0.25 pp.	+0.01	+0.04 GRC	+0.01	+0.04 GRC	0.00	+0.03 GRC
	+0.25 pp.	−0.01	−0.04 GRC	−0.01	−0.04 GRC	−0.01	−0.04 PRT
2. Model parameters							
Fiscal multiplier	0.5	−0.01	−0.07 SVK	−0.04	−0.25 POL	−0.13	−1.00 FIN
	1	+0.01	+0.07 SVK	+0.06	+0.38 FRA	+0.22	+1.64 FIN
	1.5	+0.04	+0.22 SVK	+0.28	+1.77 FRA	+0.95	+8.56 AUT
Multiplier persistence	2 years	−0.01	−0.05 SVK	−0.03	−0.18 POL	−0.08	−0.64 FIN
	4 years	+0.01	+0.06 SVK	+0.04	+0.24 FRA	+0.10	+0.68 FIN
	5 years	+0.03	+0.13 FIN	+0.08	+0.45 FRA	+0.18	+1.20 FIN
Budget semi-elasticity	×0.9	−0.01	−0.03 DEU	−0.01	−0.08 POL	−0.05	−0.40 FIN
	×1.1	+0.01	+0.03 DEU	+0.01	+0.08 POL	+0.06	+0.40 FIN

Source: Author's calculations. Note: Mean of the change across the 21 countries with adjustment requirements, and the largest change in absolute terms among them with its country. Own criterion: SPB required by the criterion the check acts on (Section 2). Runs in which the debt sustainability safeguard cannot be met are left out of the binding target. These are the runs with a multiplier of 1.5 for Finland and Spain.

3.2 Individual variation of inputs

A country has an adjustment requirement if its baseline binding target lies above its SPB in 2024. This holds for 21 countries. All aggregate statistics about changes to SPB targets across countries in this paper refer to these 21 countries. Tables 3a and 3b report the change in the SPB relative to the baseline at the three levels. Table 3a covers the individual checks of forecasts and model parameters. Table 3b covers the stochastic analysis, the alternative settings of risk tolerance and the repayment profile. For each level, the tables show the mean across the 21 countries with adjustment requirements. They also show the largest change among these countries, with the country concerned. Tables A2a and A2b in the Annex report the same for a seven-year adjustment period.

Among the forecasts, potential growth and the long-term interest rate have the largest effects. Potential growth 1 percentage point lower in $T + 10$ raises the DSA target by 0.42 percentage points on average. It raises the target by 1.21 points for Italy, 1.05 for France and 0.90 for Greece.

Table 3b: Change of the SPB relative to the baseline: estimation of risk, risk tolerance and data, four-year adjustment (percentage points of GDP)

		Own criterion		DSA target		Binding target	
Input	Value	Mean	Largest	Mean	Largest	Mean	Largest
3. Estimation of risk							
Start of stochastic projection	first adjustment year	+0.20	+0.87 GRC	0.00	+0.09 FIN	0.00	0.00
Horizon of stochastic projection	10 years	+0.14	+0.69 GRC	−0.01	−0.11 FIN	0.00	0.00
Start of shock sample	1990	−0.02	−0.08 GRC	0.00	0.00	0.00	0.00
	2010	+0.01	+0.35 SVN	0.00	+0.09 FIN	0.00	0.00
Shock frequency	annual	−0.14	−0.55 MLT	0.00	+0.07 FIN	0.00	0.00
Shock distribution	VAR bootstrap	+0.06	+0.70 GRC	+0.02	+0.46 FIN	0.00	0.00
Winsorised shocks	P1 and P99	+0.52	+1.11 ESP	+0.04	+0.36 FRA	+0.02	+0.36 FRA
	off	+0.83	+1.70 ESP	+0.09	+0.63 AUT	+0.04	+0.64 FRA
4. Risk tolerance							
Threshold, start, horizon	60%, after adj., 5 years	−0.49	−0.77 GRC	−0.01	−0.11 FIN	0.00	0.00
	80%, after adj., 5 years	+0.58	+1.05 GRC	+0.03	+0.46 FIN	+0.01	+0.08 FRA
	60%, after adj., 10 years	−0.26	−0.93 FIN	−0.01	−0.11 FIN	0.00	0.00
	80%, after adj., 10 years	+0.59	+1.61 GRC	+0.03	+0.58 GRC	+0.02	+0.45 GRC
	60%, first adj. year, 5 years	−0.42	−0.71 MLT	−0.01	−0.11 FIN	0.00	0.00
	80%, first adj. year, 5 years	+0.93	+2.51 GRC	+0.15	+1.48 GRC	+0.09	+1.23 GRC
	60%, first adj. year, 10 years	−0.17	−0.91 FIN	−0.01	−0.11 FIN	0.00	0.00
	70%, first adj. year, 10 years	+0.32	+1.56 GRC	+0.02	+0.54 GRC	+0.02	+0.39 GRC
	80%, first adj. year, 10 years	+0.89	+3.03 GRC	+0.13	+2.00 GRC	+0.11	+1.75 GRC
Adverse $r - g$ shock	0.25 pp.	−0.28	−0.50 ITA	−0.06	−0.27 GRC	−0.05	−0.28 BEL
	0.75 pp.	+0.28	+0.49 ITA	+0.16	+0.49 ITA	+0.12	+0.48 ITA
Lower-SPB shock	0.25 pp.	−0.23	−0.28 FIN	0.00	−0.05 SVK	−0.01	−0.08 SVN
	0.75 pp.	+0.23	+0.28 FIN	+0.06	+0.19 SVN	+0.04	+0.20 SVK
Financial stress shock	0.5 pp.	−0.02	−0.06 FRA	0.00	0.00	0.00	0.00
	1.5 pp.	+0.02	+0.10 FRA	0.00	0.00	0.00	0.00
Ageing cost period	5 years	−0.22	−0.93 ESP	−0.24	−0.89 ESP	−0.20	−0.88 ESP
All settings	lenient	−0.22	−0.93 ESP	−0.40	−1.25 ESP	−0.27	−1.28 ESP
	strict	0.00	0.00	+0.18	+0.49 ITA	+0.13	+0.48 ITA
Data							
Repayment profile	Eurostat profile	−0.01	−0.46 GRC	−0.01	−0.57 GRC	0.00	−0.42 GRC

Source: Author's calculations. Note: Mean of the change across the 21 countries with adjustment requirements, and the largest change in absolute terms among them with its country. Own criterion: SPB required by the criterion the check acts on (Section 2). Runs in which the debt sustainability safeguard cannot be met are left out of the binding target. The own criterion of the settings of the stochastic analysis and of the stress tests applies only to countries with a reference trajectory, and its statistics cover the 15 of them with adjustment requirements. Threshold, start, horizon: probability threshold of the stochastic criterion, start of the stochastic projection (after the adjustment period or in its first year) and horizon after the adjustment period. The baseline has a threshold of 70 percent, a start after the adjustment period, a five-year horizon, shocks of 0.5, 1 and 0.5 percentage points in the adverse $r - g$, financial stress and lower-SPB tests and an ageing cost period of ten years. Lenient: threshold of 60 percent, shocks of 0.25, 0.5 and 0.25 points and ageing cost period of five years. Strict: threshold of 80 percent and shocks of 0.75, 1.5 and 0.75 points. Hungary, Luxembourg and the Netherlands are left out of the repayment profile check for lack of data.

Growth 1 point higher lowers the target by only 0.29 points on average and by at most 0.65 points, for Spain. The SPB that the baseline debt criterion requires responds almost symmetrically. It rises by 0.54 points and falls by 0.51 points. The favourable shift moves the target less because it soon makes another criterion bind, often the deficit criterion.

A real long-term rate 1 point higher in $T + 10$ raises the DSA target by 0.23 percentage points on average. It raises the target by more than 0.1 points in 18 of the 21 countries, and most for Greece (0.47). A rate 1 point lower reduces the target by 0.23 points on average. The short-term rate has almost no effect.

With real rates unchanged, inflation affects the targets of few countries. Inflation 0.75 points

lower raises the targets of Italy and France by 0.45 and 0.43 percentage points. It raises the target of Greece by 0.30 points. In these countries, lower inflation slows the erosion of debt issued in the past at fixed rates.

Ageing costs and the long-run anchors at $T + 30$ have small effects. A shift in ageing costs of 0.2 percentage points of GDP by 2038 changes the DSA target by 0.08 points on average. It changes no target by more than 0.16 points. The scenarios of the Ageing Report differ little from its baseline by 2038 (Table A1). The long-run anchors change no target by more than 0.04 points.

The model parameters change the DSA target little on average. A multiplier of 1.0 raises it by 0.06 percentage points on average and by at most 0.38 points. In the crisis case of 1.5, the DSA target rises by 0.28 points on average. The effect is concentrated in France (1.77), Poland (1.11), Romania (0.86) and Italy (0.75). In these countries, the deficit criterion sets the target or comes close to it. A larger multiplier widens the output gap during the adjustment. The wider gap raises the cyclical deficit and thus the SPB needed to keep the deficit below 3 percent of GDP. The persistence of the multiplier and the semi-elasticity change the DSA target by less than 0.1 points on average.

3.3 Fiscal multiplier and debt sustainability safeguard

The multiplier has larger effects on the binding target. A multiplier of 1.0 raises the binding target of Finland by 1.6 points. It raises that of Austria by 1.0 points and that of Germany by 0.8 points. A persistence of five years raises the binding target of Finland by 1.2 points. A multiplier of 1.5 raises the binding target of Austria by 8.6 percentage points and that of Germany by 4.1 points. With this multiplier, Finland and Spain cannot meet the debt sustainability safeguard even with an annual adjustment of 5 percentage points. The model searches annual adjustments of up to 5 points in all runs. The mean change of the binding target in Table 3a leaves out these two countries. It therefore understates the effect.

The safeguard tests the average decline in the debt ratio during the adjustment period. A larger multiplier slows this decline. The adjustment lowers nominal GDP, and a larger adjustment lowers GDP further. With a large and persistent multiplier, additional adjustment may therefore fail to reduce the debt ratio fast enough. We describe this mechanism in Heimberger et al. (2024).

A simple calculation shows the size of this effect (see also Boussard et al., 2012; Eyraud and Weber, 2013). It ignores interest, inflation and the fading of the output effect. Let m be the multiplier, ε the budget semi-elasticity and b the debt ratio as a fraction of GDP. An increase in the SPB by x percentage points lowers the debt ratio by about $x(1 - \varepsilon m)$ through the balance. It raises the debt ratio by about bmx through lower GDP. On impact, the debt ratio therefore falls only if $m(b + \varepsilon) < 1$. For a debt ratio of 80 percent and a semi-elasticity of 0.5, this condition requires a multiplier below 0.77. Over several years, the output effect fades while the lower

deficit keeps reducing debt. The safeguard can therefore still be met at higher multipliers. With a large and persistent multiplier, the output effect fades too slowly for that.

Panel (a) of Figure 2 shows the constant annual adjustment that the safeguard requires, for each value of the multiplier. It covers the four countries where the safeguard binds or cannot be met. For this figure, I extend the search to annual adjustments of up to 20 percentage points, in steps of 0.05 points. At the baseline multiplier of 0.75, the safeguard requires an annual adjustment of 0.55 to 1.35 percentage points. The requirement rises slowly up to a multiplier of about 1.2 and steeply above it. For Finland, it reaches 4.2 points a year at a multiplier of 1.44. From a multiplier of 1.45, no annual adjustment of up to 20 points meets the safeguard. For Spain, the requirement reaches 4.0 points at a multiplier of 1.48. From 1.49, no adjustment meets the safeguard. For Spain, the average decline in the debt ratio peaks below the required 1 point, however large the adjustment. For Finland, a larger adjustment lowers GDP enough to raise debt above 90 percent of GDP in the first years. The safeguard then requires a larger decline. The limit of 5 points on the annual adjustment in the main runs is therefore not the reason the safeguard cannot be met.

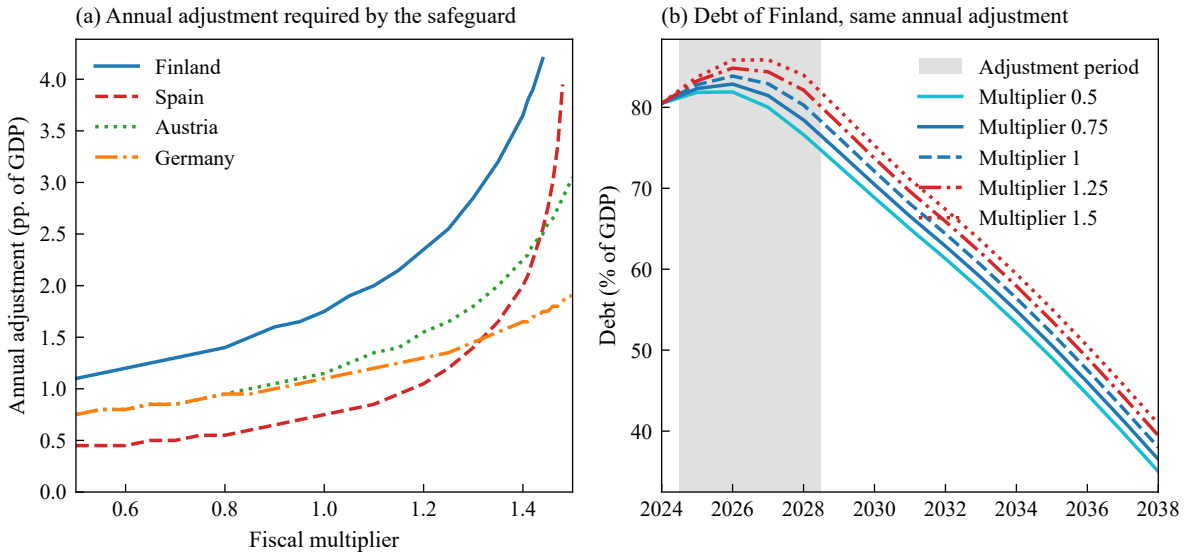


Figure 2: Debt sustainability safeguard and fiscal multiplier, four-year adjustment

Source: Author's calculations. Note: (a) Smallest constant annual adjustment, in steps of 0.05 percentage points, for which the debt sustainability safeguard is met, including the minimum adjustment of the EDP and the deficit resilience safeguard, with the persistent multiplier of the baseline, for multipliers from 0.5 to 1.5, in steps of 0.05 and of 0.01 above 1.4. Lines end where no annual adjustment of up to 20 percentage points meets the safeguard. The baseline multiplier is 0.75. (b) Debt of Finland for a constant annual adjustment of 1.34 percentage points, the adjustment of its baseline binding target. The safeguard is met with multipliers of 0.5 and 0.75 and not met with multipliers of 1.0 and more. The shaded area marks the adjustment period.

The multiplier mainly changes the timing of the debt decline. Panel (b) of Figure 2 shows the debt ratio of Finland for multipliers from 0.5 to 1.5. The annual adjustment is the same in all cases, 1.34 percentage points, the adjustment of the baseline binding target. During the adjustment, a larger multiplier slows the decline. Debt is 80.5 percent of GDP in 2024. In 2028, it ranges from 76.6 percent with a multiplier of 0.5 to 84.0 percent with a multiplier of 1.5. The

safeguard is not met with multipliers of 1.0 or more. After the adjustment, the output effect fades. Debt then falls at a similar pace in all cases, by about 4 points a year. Debt nevertheless stays higher with a larger multiplier, because the debt accumulated during the adjustment is not reversed. In 2038, debt is 40.9 percent of GDP with a multiplier of 1.5 and 35.0 percent with a multiplier of 0.5. The safeguard tests the change in debt during the adjustment, which the multiplier affects strongly. The decline after the adjustment hardly depends on the multiplier. A safeguard that measured the decline in debt over a longer period would therefore require a smaller adjustment.

The long-run effect of the multiplier on debt is therefore small. In a crisis with large adjustment needs, its short-run effect can still matter for fiscal policy. With a multiplier of 1.25 or more, Finland's debt rises to about 85 percent of GDP in the first two adjustment years. It is still higher in 2028 than in 2024. A consolidation can then add to the pressure on public finances in the short run, because the debt ratio rises despite the adjustment. Boussard et al. (2012) and Eyraud and Weber (2013) show when consolidations raise the debt ratio in the short run. Heimberger et al. (2024) discuss the consequences for the EU rules. Evidence from the European consolidations after 2010 links multipliers of this size to two conditions (Blanchard and Leigh, 2013; House et al., 2020). Interest rates are at their lower bound, and many countries consolidate at the same time. Consolidations also have larger output costs in downturns (Jordà and Taylor, 2016). They can lower output for a long time (Fatás and Summers, 2018; Gechert et al., 2019). Other studies find smaller costs, in particular for consolidations based on spending cuts (Guajardo et al., 2014; Alesina et al., 2019). In the joint draws, the multiplier stays within the range for normal times, and the safeguard can be met in all draws. A safeguard that cannot be met is therefore a risk under crisis conditions only.

For Finland, the safeguard binds partly because of the stock-flow adjustments that the Commission projects. Stock-flow adjustments are changes in debt that the deficit does not explain. In Finland, they reflect surpluses of pension funds. These surpluses lower the deficit but not gross debt (Boivin and Darvas, 2025). Gross debt therefore falls more slowly than the deficit suggests. I keep these projections in all analyses, like all other country-specific inputs of the Commission.

Table A6 in the Annex repeats the multiplier checks with the Commission's rule for closing the output gap. All effects have the same sign as with the persistent multiplier and are larger. With a multiplier of 1.5, Austria also cannot meet the safeguard. The binding target of Germany rises by 7.00 instead of 4.08 percentage points. The safeguard results therefore do not depend on the specification of the multiplier in the baseline.

3.4 Joint variation of forecasts and model parameters

Individual checks miss interactions between inputs. For example, a large multiplier matters more when its effect is persistent. Adverse forecasts also add up once several criteria bind. I

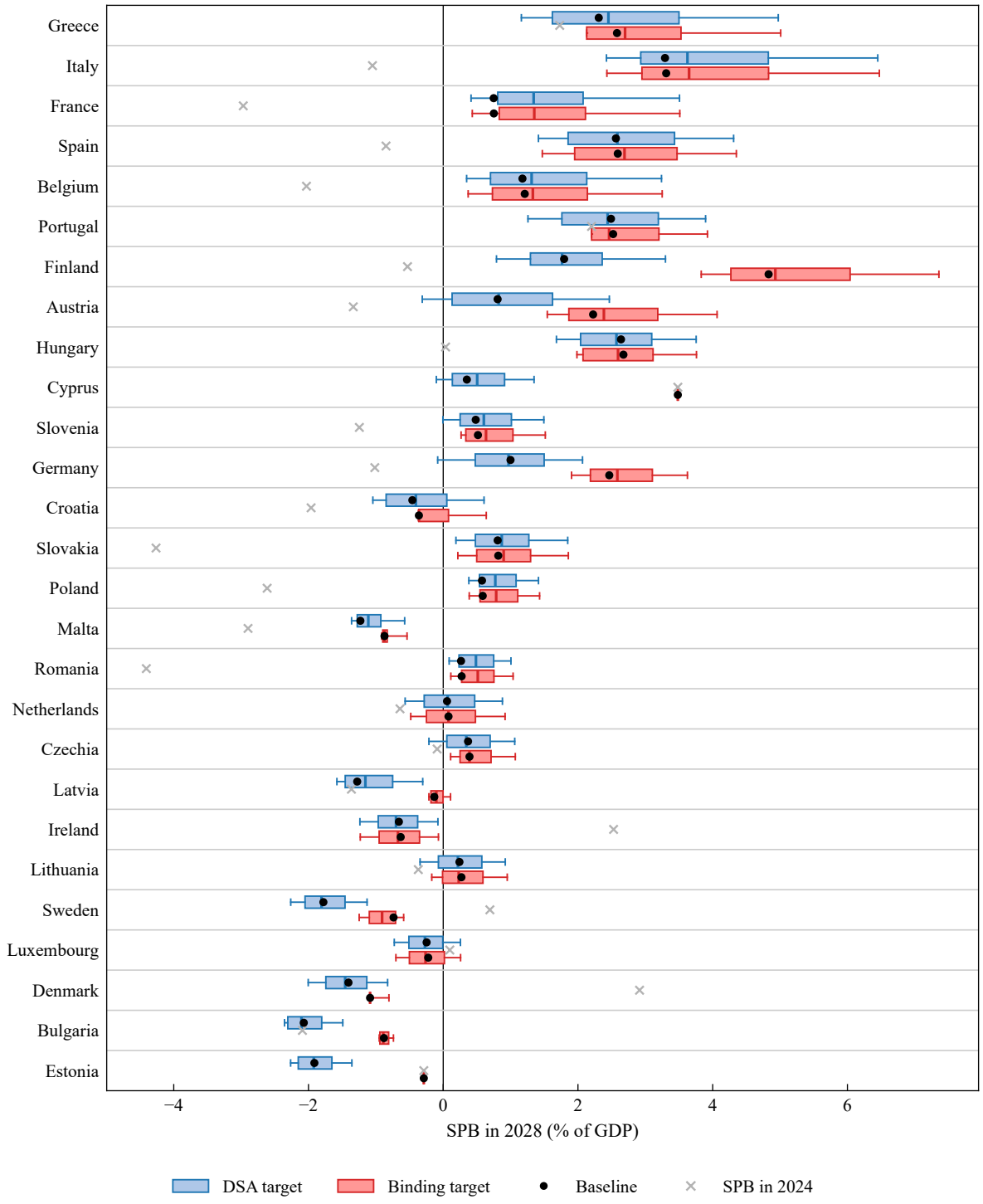


Figure 3: SPB targets by country: baseline and joint draws, four-year adjustment (percent of GDP)

Source: Author's calculations. Note: SPB at the end of the adjustment period. For each country, boxes show the 25th to 75th percentiles of the DSA target (blue) and of the binding target (red) across 500 joint draws of forecasts and model parameters, lines in the boxes the median and whiskers the 10th to 90th percentiles. Dots show the baseline targets and grey crosses the SPB in 2024. Countries are ordered by debt in 2024.

therefore draw the forecasts and the model parameters jointly in 500 draws. Latin hypercube sampling spreads the draws evenly over their distributions. The forecast errors are drawn from their empirical distribution. Errors from the same source are drawn together. The short- and

long-term rate errors of the same year form one pair. The growth and inflation errors of the same country and vintage form another. Errors from different sources are drawn independently, since the sources cover different samples and horizons. The samples overlap over the years 2000 to 2015. Over these years, growth and rate errors are positively correlated. Independent draws therefore somewhat overstate the dispersion of $r - g$. The multiplier is drawn from a uniform distribution between 0.5 and 1.0. The semi-elasticity is drawn from a uniform distribution over its range. The persistence of the multiplier takes each value from two to five years with equal probability. The settings of the third and fourth groups stay at their baseline values. Each draw is a full model run for the four-year adjustment period, including the stochastic criterion.

The draws give the distribution of targets implied by past forecast errors and by the range of published parameter estimates. I report the median and the 25th and 75th percentiles of the change in the target across draws. I call the 25th percentile the favourable combination of assumptions and the 75th percentile the adverse combination. These percentiles cover the middle half of the outcomes. They keep the tails of the error distributions from driving the results. The tails include, for example, the growth errors of the boom-bust years 2006 to 2008 in Ireland, the Baltic countries and Greece.

Figure 3 and Table 4 show the results by country. Figure A1 in the Annex shows the distribution of debt across the draws when the SPB follows the path of the baseline binding target. The average required adjustment of the 21 countries with adjustment requirements is 2.6 percentage points over four years. Six of these countries receive only technical information. For these 21 countries, the favourable combination lowers the DSA target by 0.36 points on average. The median draw raises it by 0.09 points, and the adverse combination raises it by 0.65 points. The favourable combination lowers the binding target by 0.26 points. The median draw raises it by 0.10 points, and the adverse combination by 0.63 points. Per year of the adjustment, the favourable and adverse changes in the DSA target amount to about 0.09 and 0.16 points.

Under the rules, a control account records the deviations of net expenditure from its agreed path. The rules tolerate deviations of up to 0.3 percentage points of GDP in a year and 0.6 points cumulatively. For countries with debt above 60 percent of GDP, larger deviations lead to a Commission report under Article 126(3) of the Treaty (Council of the European Union, 2024). The adverse change in the target over four years is about as large as the cumulative deviation that the rules tolerate. The comparison is approximate. Net expenditure follows from the SPB path but is not identical to it.

The adverse side is larger because the target rises without limit under adverse inputs. Favourable inputs instead soon make another criterion bind, and for the binding target also the adjustment floor. The error distributions do not explain this asymmetry. Between the 25th and 75th percentiles, the rate and growth errors are, if anything, larger on the favourable side (Table A1). Large adverse rate errors appear only in the upper tail, at the 90th percentile.

The DSA target moves most for Italy, from 0.36 points below to 1.53 points above the baseline. The ranges of Greece, Spain, Austria, Portugal and Belgium span between 1.4 and

1.9 points. The effects are most asymmetric where the deficit criterion sets the target or lies close to it, as in France, Poland and Romania. In these countries, a larger multiplier raises the SPB that the deficit criterion requires. Favourable forecasts soon make another criterion bind. For France, even the favourable combination raises the DSA target, by 0.06 points. The adverse combination raises it by 1.33 points. In the other countries, the results mainly reflect the forecasts. A multiplier of 1.0 moves the DSA target by more than 0.1 points in only four countries.

For the binding target, the debt sustainability safeguard adds to the effects of the adverse combination. The adverse combination raises the binding target of Finland by 1.21 percentage points, against 0.56 points for its DSA target. It raises the binding target of Austria by 0.96 points and that of Germany by 0.64 points. The safeguard can be met in all draws. In 11 percent of all country-draws, the binding target requires a total adjustment of more than 5 percentage points over four years. Adjustments of this size have been rare. Darvas et al. (2025a) study 25 EU countries and 11 other advanced economies between 1970 and 2022. Improvements in the primary balance of more than 5 points of GDP occurred in fewer than 10 percent of periods of up to seven years. The share of draws with such adjustments is 66 percent for Finland and 59 percent for Slovakia. It is 43 percent for Italy, 37 percent for Romania and 26 percent for France. For Finland and Slovakia, the baseline target already requires an adjustment of more than 5 points.

4 Stochastic analysis, stress tests and risk tolerance

4.1 Specification of the stochastic analysis

The stochastic criterion depends on choices about how the DSA estimates risk. These are modelling choices that should be made on statistical grounds. I vary six of these choices one at a time. First, the simulated shocks start in the first year of the adjustment period instead of the year after it. Shocks to the primary balance are set to zero during the adjustment period. The criterion itself is unchanged. Debt five years after the adjustment period must be below debt at the end of the adjustment period. The shocks of the adjustment years therefore add to the uncertainty. Second, the horizon of the stochastic projection is ten instead of five years after the adjustment period. Third, the shock sample starts in 1990 or 2010 instead of 2000. Fourth, shocks are annual instead of quarterly. Fifth, shocks are drawn by bootstrap from a vector autoregression (VAR), as in Darvas et al. (2025a), instead of from a normal distribution. Sixth, the historical shocks are winsorised at the 1st and 99th percentiles, or not at all, instead of at the 5th and 95th percentiles. The Commission plans to allow for persistent shocks and to replace the normal distribution of shocks with a bootstrap (Bec et al., 2025). The VAR bootstrap addresses the same two points. The start and the horizon of the stochastic projection are not part of these planned refinements.

Table 4: SPB targets by country: baseline and joint draws of forecasts and model parameters, four-year adjustment (percent of GDP and percentage points)

			Own	DSA target					Binding target				
	Debt	Required	Band	Base	P25	Median	P75	Band	Base	P25	Median	P75	Band
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
Greece	153.9	0.85	2.48	2.31	−0.69	+0.14	+1.19	1.88	2.58	−0.45	+0.12	+0.95	1.40
Italy	138.6	4.36	2.79	3.29	−0.36	+0.33	+1.53	1.89	3.31	−0.36	+0.34	+1.52	1.88
France	112.4	3.72	1.89	0.75	+0.06	+0.59	+1.33	1.27	0.75	+0.08	+0.60	+1.36	1.28
Spain	105.5	3.44	1.93	2.56	−0.71	+0.02	+0.87	1.58	2.59	−0.64	+0.10	+0.88	1.52
Belgium	103.4	3.24	2.08	1.18	−0.47	+0.14	+0.95	1.43	1.21	−0.48	+0.12	+0.93	1.41
Portugal	95.6	0.32	1.54	2.49	−0.73	−0.05	+0.70	1.43	2.52	−0.32	−0.06	+0.68	1.00
Finland	80.5	5.36	1.05	1.79	−0.50	−0.03	+0.56	1.07	4.83	−0.56	+0.10	+1.21	1.77
Austria	79.5	3.56	1.67	0.81	−0.68	+0.02	+0.82	1.49	2.22	−0.36	+0.16	+0.96	1.32
Hungary	74.3	2.64	1.32	2.64	−0.60	−0.07	+0.46	1.06	2.67	−0.60	−0.08	+0.44	1.04
Cyprus	70.6	0.00	0.95	0.35	−0.22	+0.15	+0.56	0.78	3.48	0.00	0.00	0.00	0.00
Slovenia	68.1	1.76	0.88	0.48	−0.23	+0.12	+0.53	0.76	0.52	−0.18	+0.12	+0.52	0.70
Germany	62.5	3.48	1.02	1.00	−0.52	−0.03	+0.50	1.02	2.46	−0.28	+0.12	+0.64	0.92
Croatia	59.5	1.60	0.97	−0.46	−0.39	+0.05	+0.51	0.90	−0.36	0.00	0.00	+0.44	0.44
Slovakia	58.5	5.08	0.89	0.81	−0.33	+0.06	+0.46	0.79	0.82	−0.32	+0.08	+0.48	0.80
Poland	53.7	3.20	0.96	0.58	−0.04	+0.20	+0.51	0.54	0.59	−0.04	+0.20	+0.52	0.56
Malta	52.0	2.03	0.93	−1.23	−0.05	+0.12	+0.30	0.35	−0.87	−0.03	0.00	+0.04	0.07
Romania	50.9	4.68	0.91	0.26	−0.03	+0.22	+0.48	0.52	0.27	0.00	+0.24	+0.48	0.48
Netherlands	47.1	0.72	0.90	0.06	−0.34	+0.01	+0.41	0.75	0.08	−0.33	0.00	+0.40	0.73
Czechia	45.2	0.48	0.83	0.37	−0.32	−0.03	+0.33	0.64	0.39	−0.14	0.00	+0.32	0.46
Latvia	44.5	1.23	0.89	−1.28	−0.18	+0.12	+0.53	0.70	−0.13	−0.05	+0.03	+0.13	0.18
Ireland	42.5	−3.16	0.87	−0.66	−0.31	−0.04	+0.28	0.59	−0.63	−0.32	−0.04	+0.28	0.60
Lithuania	38.3	0.64	0.81	0.24	−0.31	−0.02	+0.33	0.64	0.27	−0.28	−0.04	+0.32	0.60
Sweden	32.0	−1.43	0.85	−1.78	−0.27	−0.03	+0.32	0.59	−0.74	−0.36	−0.17	+0.03	0.39
Luxembourg	27.1	−0.32	0.80	−0.25	−0.26	−0.03	+0.24	0.50	−0.22	−0.28	−0.04	+0.24	0.52
Denmark	26.5	−4.00	0.87	−1.41	−0.34	−0.05	+0.27	0.61	−1.09	0.00	0.00	0.00	0.00
Bulgaria	24.5	1.21	0.74	−2.07	−0.24	−0.03	+0.27	0.50	−0.88	−0.06	−0.01	+0.07	0.13
Estonia	21.4	0.00	0.76	−1.91	−0.24	−0.01	+0.26	0.50	−0.29	0.00	0.00	0.00	0.00
Mean (adjusting)	73.7	2.55	1.31	0.79	−0.36	+0.09	+0.65	1.01	1.23	−0.26	+0.10	+0.63	0.89

Source: Author's calculations. Note: Column (1) is debt in 2024. Column (2) is the adjustment that the baseline binding target requires, relative to the SPB in 2024. Column (3) is the band of the SPB required by the own criterion: the debt criterion of the baseline scenario. Base is the baseline target. P25, Median and P75 are the 25th, 50th and 75th percentiles of the change of the target across 500 joint draws, relative to the baseline. Band is the difference between P75 and P25. The last row gives the mean across the 21 countries whose binding target requires a positive adjustment.

These choices move the SPB that the stochastic criterion requires. With winsorisation at the 1st and 99th percentiles, this SPB rises by 0.52 percentage points on average. Without winsorisation, it rises by 0.83 points, and with an early start by 0.20 points. With annual shocks, it falls by 0.14 points. Most of these changes do not pass through to the target. In the baseline, the stochastic criterion sets the DSA target only for Finland. For all other countries, it lies below other criteria.

The winsorisation has the largest effect on the targets. Winsorisation at the 1st and 99th percentiles raises the DSA target of France by 0.36 percentage points. It raises those of Finland and Austria by 0.27 and 0.24 points. Without winsorisation, the targets of France and Austria rise by 0.63 points. Those of Finland and Italy rise by 0.38 and 0.17 points. The VAR bootstrap raises the target of Finland by 0.46 points. The other choices change no target by more than 0.11 points.

The early start adds the shocks of the adjustment years to the uncertainty. The criterion, however, still compares debt after the adjustment period with debt at its end. The SPB that

the criterion requires therefore rises by only 0.20 points on average, and no target changes by more than 0.1 points. Combined with a ten-year horizon, the early start raises the target of Greece by 0.54 points. Wieland (2024) and Erce (2025) propose to include uncertainty during the adjustment period. On their own, the shocks of the adjustment years change the targets little. The start and the horizon matter more in combination with the probability threshold, as Section 4.2 shows.

4.2 Calibration of risk tolerance

Three settings determine how much uncertainty the rules accept: the probability threshold, the size of the stress tests and the ageing cost period. The ageing cost period is the number of years after the adjustment period over which changes in ageing costs enter the projection. I refer to these settings together as risk tolerance. They have no true value that data could reveal. Blanchard et al. (2021) make a similar point in their proposal to base the EU fiscal rules on a stochastic DSA. I therefore do not assign the settings a range. I report instead how the SPB changes relative to the baseline under alternative settings. Table 3b reports these changes at the three levels. For the stochastic criterion, the table shows a grid of threshold, start and horizon. The stress tests and the ageing cost period vary one at a time. A lenient and a strict scenario combine the alternative settings at each end. The notes to Table 3b list their settings.

The probability threshold has a large effect on the stochastic criterion itself. A threshold of 80 percent raises the SPB that this criterion requires by 0.58 percentage points on average. The increase exceeds 0.1 points in all 15 countries with a reference trajectory and adjustment requirements. A threshold of 60 percent lowers the SPB by 0.49 points. Neither choice passes through much to the targets, however. With a threshold of 80 percent, the DSA targets rise by 0.03 points on average. Only those of Finland and Austria rise by more than 0.1 points. The threshold has a larger effect in combination with an early start. With an early start and a five-year horizon, a threshold of 80 percent raises the DSA target by 0.15 points on average. It raises the target of Greece by 1.48 points and that of Finland by 0.60 points. With a ten-year horizon, it raises the target of Greece by 2.0 points.

Among the stress tests, the size of the adverse $r - g$ test matters most. In this test, interest rates rise and growth falls by 0.5 percentage points each. Shocks of 0.75 points raise the SPB that the test requires by 0.28 points on average. They raise the DSA target by 0.16 points and the binding target by 0.12 points. The largest effect on the DSA target is 0.49 points, for Italy. Shocks of 0.25 points lower the DSA target by 0.06 points on average. A lower-SPB shock of 0.75 points raises the DSA target by 0.06 points. The size of the financial stress test changes no target.

The size of the stress tests has no true value. Past forecast errors can still show how much uncertainty the stress tests accept. The adverse $r - g$ test raises $r - g$ by 1 percentage point. I compare this shift with the error in the long-term rate minus the error in growth, using the errors

of Section 3. This difference exceeded 1 point in 30 percent of cases. The share is the same when the two errors are combined independently and when they are paired by year for 2000 to 2015. The test therefore covers past errors in $r - g$ up to about their 70th percentile. This percentile is close to the probability threshold of 70 percent in the stochastic criterion. The two numbers refer to different quantities, however. The percentile refers to an error in an input. The threshold refers to a probability about the debt outcome. The two also refer to different sources of risk. The stress test covers errors in the central projection, and the stochastic criterion covers future shocks around it. Shocks of 0.75 points shift $r - g$ by 1.5 points and cover errors up to the 75th percentile. The comparison is approximate. The growth errors are averages over five years, whereas the rate errors refer to rates ten years ahead.

After the adjustment period, the DSA keeps the SPB before ageing costs at its target. Ageing costs are public spending on pensions, health care and long-term care. Any increase in these costs relative to the last adjustment year then lowers the SPB one for one. Any decrease raises it (Darvas et al., 2025b). The baseline ageing cost period of ten years covers the whole DSA horizon. It assumes that governments adjust no other part of the budget to offset changes in ageing costs. For countries whose ageing costs rise, the rules thus accept no risk that such an offset fails. For countries whose ageing costs fall, the ten-year period instead credits them with the full decline. With a period of five years, the DSA counts only the changes in ageing costs up to the fifth year.

The ageing cost period changes the targets of most countries. A period of five instead of ten years lowers the DSA target by 0.24 percentage points on average. It lowers the target of Spain by 0.89 points. It raises the targets of Bulgaria, Latvia, Estonia, Croatia and Sweden, whose ageing costs fall. For these countries, the shorter period is the stricter setting. The lenient scenario lowers the DSA target by 0.40 points on average, mostly through the ageing cost period. The strict scenario raises it by 0.18 points. The two scenarios are asymmetric because the ageing cost period cannot be longer than the DSA horizon. Table A3 in the Annex reports the effects by country.

Table 5 compares the effects of the settings of risk tolerance with those of the forecasts and model parameters. The lenient scenario lowers the DSA target by 0.40 points on average. The favourable combination of the joint draws lowers it by 0.36 points. Adverse $r - g$ shocks of 0.75 instead of 0.5 percentage points raise the DSA target by 0.16 points. This increase is a quarter of the adverse combination. The strict scenario raises the DSA target by only 0.18 points, far less than the adverse combination. The settings of risk tolerance thus match the effect of the forecasts mainly on the lenient side. The lenient result comes mostly from the shorter ageing cost period. The range of the forecasts comes from data, whereas the alternative settings of risk tolerance are values that I chose. The specification of the stochastic analysis matters much less. The VAR bootstrap raises the DSA target by only 0.02 points.

The current settings imply that the rules protect against errors in $r - g$ up to about their 70th percentile. They also imply that governments offset none of the change in ageing costs over the

Table 5: Uncertainty about the economy and tolerance for risk: mean change of the targets, four-year adjustment (percentage points of GDP)

Setting	DSA target	Binding target
Forecasts and model parameters		
Joint draws, 25th percentile (favourable)	−0.36	−0.26
Joint draws, 75th percentile (adverse)	+0.65	+0.63
Potential growth T + 10, −1 pp.	+0.42	+0.35
Real long-term rate T + 10, +1 pp.	+0.23	+0.17
Fiscal multiplier 1.0	+0.06	+0.22
Specification of the stochastic analysis		
VAR bootstrap	+0.02	0.00
No winsorisation	+0.09	+0.04
Start in first adjustment year	0.00	0.00
Risk tolerance		
Probability threshold 60%	−0.01	0.00
Probability threshold 80%	+0.03	+0.01
Threshold 80%, start in first adjustment year	+0.15	+0.09
Adverse $r - g$ shocks 0.25 pp.	−0.06	−0.05
Adverse $r - g$ shocks 0.75 pp.	+0.16	+0.12
Lower-SPB shock 0.25 pp.	0.00	−0.01
Lower-SPB shock 0.75 pp.	+0.06	+0.04
Ageing cost period 5 years	−0.24	−0.20
Lenient scenario	−0.40	−0.27
Strict scenario	+0.18	+0.13

Source: Author's calculations. Note: Mean change of the target relative to the baseline across the 21 countries with adjustment requirements. Divide by four for the change in the annual adjustment. Joint draws: 25th and 75th percentiles of the change across 500 draws of forecasts and model parameters. The other rows are individual checks and alternative settings from Tables 3a and 3b. Runs in which the debt sustainability safeguard cannot be met are left out of the binding target.

DSA horizon. These levels are set by convention. Whether they are appropriate is a question for policymakers.

5 Debt repayment profile and data vintage

The repayment profile of debt matters mostly for Greece and Finland. In the baseline, the share of long-term debt that matures each year follows the Commission's inputs. These inputs converge to a historical average. The alternative repays the debt outstanding in 2024 according to Eurostat data on debt by residual maturity. These data are not available for Hungary, Luxembourg and the Netherlands. Maturing debt is refinanced at market rates. In Greece and Finland, market rates exceed the average interest rate on the existing debt. The repayment profile therefore determines how fast the average interest rate rises.

In the Commission's inputs for Greece, the share of long-term debt that matures each year is about 4 percent in 2026. It rises to between 9 and 11 percent after 2033. In the Eurostat data, this share stays between 2 and 4 percent. Much of Greek debt consists of official loans with long maturities. Less debt is therefore refinanced at market rates. For the same SPB path, the

average interest rate in 2038 is 3.0 instead of 3.5 percent. Debt is 3 percentage points of GDP lower. The SPB that the adverse $r - g$ stress test requires falls accordingly. The DSA target of Greece falls by 0.57 points and its binding target by 0.42 points.

For Finland, the effect goes in the opposite direction. The Eurostat data show about 14 percent of long-term debt maturing each year until 2028, against 3 to 4 percent in the Commission's inputs. More debt is refinanced at market rates. The average interest rate in 2038 is 2.7 instead of 2.2 percent, and debt is 3 points higher. The SPB that the stochastic criterion requires rises, and with it the DSA target, by 0.26 points. The binding target, which the debt sustainability safeguard sets, rises by 0.32 points. The targets of the other countries change by less than 0.15 points.

The sensitivity analysis so far has used the 2024 data of the Commission's prior guidance. Later applications of the rules will use newer data. I therefore rerun the individual checks and the alternative settings of risk tolerance with the data from September 2026. The four-year adjustment period then runs from 2027 to 2030. The targets themselves differ from those with the 2024 data (Table A5 in the Annex). The effects of the inputs hardly change. Table A4 in the Annex compares the mean effect of each check across the two vintages. The correlation is 0.99 for the DSA target and 0.98 for the binding target. On average, the mean effects differ by about 0.02 percentage points. The largest differences concern potential growth and the crisis multiplier. With the 2024 data, potential growth 1 point higher lowers the DSA target by 0.29 points. With the 2026 data, it lowers the target by 0.35 points. A multiplier of 1.5 raises the binding target by 0.95 and 0.82 points with the two vintages. The sensitivities reported in this paper are therefore not specific to the conditions of 2024.

The average effects hide changes for individual countries. With the newer data, the criterion that sets the target changes for several countries. A different DSA criterion sets the DSA target for four countries. A different constraint sets the binding target for eight (Tables 1 and A5). The debt sustainability safeguard no longer binds for Austria. Its binding target falls from 2.22 to 1.16 percent of GDP. Finland's binding target rises from 4.83 to 8.49 percent of GDP, which requires an annual adjustment of 2.45 percentage points. Which inputs matter for a country therefore changes when its binding criterion changes.

6 Conclusions

The effect of an input of the DSA on the SPB targets depends on which criterion binds. Forecasts and model parameters shift all deterministic criteria at once, so they matter for most countries. I draw forecasts and model parameters jointly from past forecast errors and published estimates. At the 25th percentile of these draws, the DSA target of the countries with adjustment requirements falls by 0.36 percentage points on average. At the 75th percentile, it rises by 0.65 points. The average required adjustment is 2.6 points over four years. The adverse change is about as large as the cumulative deviation that the control account tolerates. The draws use errors from which

I removed the average. They therefore measure the size of forecast errors but not their bias. Growth forecasts, for instance, have been optimistic on average in the past (Frankel and Schreger, 2013).

Within the alternatives I consider, modelling choices reflecting the tolerance for risk move the targets by similar amounts as forecast errors. This holds mainly on the lenient side. The lenient scenario lowers the DSA target by 0.40 points. Adverse $r - g$ shocks of 0.75 instead of 0.5 percentage points raise it by 0.16 points. The specification of the stochastic analysis matters much less, because the stochastic criterion sets the target only for Finland. The refinements that the Commission plans for the stochastic analysis would therefore change few targets (Bec et al., 2025). As a result, the rules protect against errors in the central projection mainly through the deterministic stress tests. The adverse $r - g$ test, for example, is larger than about 70 percent of past forecast errors, implying an acceptable degree of uncertainty about the central projection similar to the probability threshold of the stochastic DSA.

These results suggest three priorities for the review of the DSA. First, the size of the stress tests and the ageing cost period deserve as much attention as the stochastic simulations. The stochastic approach is more coherent in theory, but it currently sets the target for only one country. Second, the debt sustainability safeguard can interact with the fiscal multiplier and become self-defeating, because a larger adjustment slows the decline in debt that it tests. Measuring the decline of debt over a longer window would reduce this effect, and dropping the safeguard would remove it. Third, the level of risk that the rules implicitly accept via stress tests and probability thresholds should be justified explicitly.

Future reviews should also consider how policies that raise potential growth could ease the targets. Potential output in the DSA follows the EU's commonly agreed methodology, which does not respond to policy. In Gottschalk et al. (2025), we let selected measures of the medium-term fiscal plans raise potential growth in the DSA. Germany's infrastructure package, for example, then lowers its SPB target from 0.99 to 0.84 percent of GDP. The DSA also leaves out the reverse channel, through which consolidations can lower potential output for a long time (Fatás and Summers, 2018; Gechert et al., 2019).

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Annex

Table A1: Distribution of past forecast errors

Assumption	Source of errors	Sample	*N*	Mean error	P10	P25	P75	P90
Real short-term rate, $T+10$	Bundesbank: 1-year rate in ten years minus forward rate	Germany, year ends 1986-2015	30	−4.13	−1.61	−1.22	+0.40	+3.11
Real long-term rate, $T+10$	Bundesbank: 10-year rate in ten years minus 10Y10Y forward rate	Germany, year ends 1986-2015	30	−2.98	−1.53	−1.24	+0.89	+2.36
Inflation, $T+10$	IMF WEO: CPI inflation, average error over five years	EU27, autumn vintages 2000-2018	513	+0.14	−1.33	−0.74	+0.64	+1.35
Potential growth, $T+10$	IMF WEO: real GDP growth, average error over five years	EU27, autumn vintages 2000-2018	513	−0.70	−2.53	−0.84	+1.08	+2.31
Change in ageing costs, 2024–2038	Ageing Report 2024: scenarios except pension policy, change in ageing costs 2024-2038	EU27, 8 scenarios	216	–	−0.38	−0.24	+0.14	+0.62

Source: Author's calculations based on Ech-charfi (2024), the Deutsche Bundesbank term structure of interest rates and European Commission (2024a). Note: Errors are realised minus forecast values, in percentage points. WEO errors are averages over the five forecast years of each autumn vintage, demeaned by country. Forward-rate errors compare the forward rate at each year end with the rate realised ten years later, demeaned. Ageing costs: deviation of each scenario from the baseline change in the total cost of ageing between 2024 and 2038, in percentage points of GDP, for all countries and all scenarios except the pension policy scenarios. Mean error: average error before demeaning. P10 to P90: percentiles of the demeaned errors. The individual checks shift each forecast by a round amount close to half the distance between the 25th and the 75th percentile; the joint draws use the full distribution.

Table A2a: Change of the SPB relative to the baseline: forecasts and model parameters, seven-year adjustment (percentage points of GDP)

		Own criterion		DSA target		Binding target	
Input	Value	Mean	Largest	Mean	Largest	Mean	Largest
1. Forecasts							
Real short-term rate, $T + 10$	−1 pp.	−0.04	−0.15 ITA	−0.04	−0.15 ITA	−0.03	−0.18 ITA
	+1 pp.	+0.04	+0.17 ITA	+0.04	+0.15 ITA	+0.04	+0.15 ITA
Real long-term rate, $T + 10$	−1 pp.	−0.30	−0.65 GRC	−0.27	−0.47 ITA	−0.17	−0.42 ESP
	+1 pp.	+0.30	+0.60 GRC	+0.30	+0.57 ITA	+0.21	+0.48 GRC
Inflation, $T + 10$	−0.75 pp.	+0.15	+0.50 ITA	+0.10	+0.47 GRC	+0.06	+0.42 GRC
	+0.75 pp.	−0.13	−0.41 ITA	+0.01	+0.30 ITA	+0.01	−0.28 FIN
Potential growth, $T + 10$	−1 pp.	+0.54	+1.34 ITA	+0.47	+1.10 ITA	+0.36	+0.86 ITA
	+1 pp.	−0.51	−1.16 ITA	−0.29	−0.60 FIN	−0.16	−0.49 FIN
Ageing costs	−0.2 pp.	−0.07	−0.15 AUT	−0.08	−0.14 AUT	−0.05	−0.14 CZE
	+0.2 pp.	+0.08	+0.15 AUT	+0.09	+0.14 AUT	+0.06	+0.14 CZE
Real short-term rate, $T + 30$	−0.5 pp.	0.00	−0.02 PRT	0.00	−0.02 PRT	0.00	0.00
	+0.5 pp.	0.00	+0.02 PRT	0.00	+0.02 PRT	0.00	0.00
Real long-term rate, $T + 30$	−0.5 pp.	−0.02	−0.07 GRC	−0.02	−0.05 AUT	0.00	−0.04 ITA
	+0.5 pp.	+0.02	+0.07 GRC	+0.02	+0.06 GRC	+0.01	+0.07 ESP
Inflation, $T + 30$	−0.25 pp.	+0.02	+0.06 GRC	+0.01	+0.05 GRC	+0.01	+0.07 PRT
	+0.25 pp.	−0.02	−0.06 GRC	−0.01	−0.05 BEL	0.00	−0.02 BEL
2. Model parameters							
Fiscal multiplier	0.5	−0.01	−0.07 SVK	−0.02	−0.06 SVK	−0.07	−0.42 FIN
	1	+0.01	+0.07 SVK	+0.03	+0.23 FRA	+0.08	+0.56 FIN
	1.5	+0.04	+0.21 SVK	+0.13	+0.79 FRA	+0.31	+2.10 FIN
Multiplier persistence	2 years	0.00	+0.13 ITA	−0.01	−0.05 FIN	−0.05	−0.28 FIN
	4 years	+0.02	+0.05 SVK	+0.03	+0.16 FRA	+0.05	+0.35 FIN
	5 years	+0.03	+0.11 FRA	+0.05	+0.33 FRA	+0.10	+0.63 FIN
Budget semi-elasticity	×0.9	−0.01	−0.02 ROU	−0.01	−0.04 FRA	−0.03	−0.14 FIN
	×1.1	+0.01	+0.02 ROU	+0.01	+0.06 FRA	+0.02	+0.21 FIN

Source: Author's calculations. Note: See Table 3a. With a seven-year adjustment, the debt sustainability safeguard can be met in all runs.

Table A2b: Change of the SPB relative to the baseline: estimation of risk, risk standard and data, seven-year adjustment (percentage points of GDP)

		Own criterion		DSA target		Binding target	
Input	Value	Mean	Largest	Mean	Largest	Mean	Largest
3. Estimation of risk							
Start of stochastic projection	first adjustment year	+0.48	+1.98 GRC	+0.14	+1.96 GRC	+0.10	+1.82 GRC
Horizon of stochastic projection	10 years	0.00	−0.80 FIN	0.00	+0.09 GRC	0.00	+0.06 GRC
Start of shock sample	1990	−0.02	−0.08 GRC	0.00	0.00	0.00	0.00
	2010	+0.01	+0.34 SVN	0.00	0.00	0.00	0.00
Shock frequency	annual	−0.14	−0.56 MLT	0.00	0.00	0.00	0.00
Shock distribution	VAR bootstrap	+0.06	−0.70 ROU	+0.06	+0.60 GRC	+0.03	+0.54 GRC
Winsorised shocks	P1 and P99	+0.52	+1.08 ESP	+0.10	+0.53 FRA	+0.04	+0.25 ITA
	off	+0.82	+1.64 ESP	+0.21	+0.99 GRC	+0.11	+0.84 GRC
4. Risk tolerance							
Threshold, start, horizon	60%, after adj., 5 years	−0.50	−0.90 GRC	0.00	0.00	0.00	0.00
	80%, after adj., 5 years	+0.58	+1.03 GRC	+0.11	+1.00 GRC	+0.06	+0.84 GRC
	60%, after adj., 10 years	−0.40	−1.09 FIN	0.00	0.00	0.00	0.00
	80%, after adj., 10 years	+0.45	+0.98 GRC	+0.07	+0.96 GRC	+0.05	+0.84 GRC
	60%, first adj. year, 5 years	−0.30	−0.71 MLT	0.00	0.00	0.00	0.00
	80%, first adj. year, 5 years	+1.37	+4.26 GRC	+0.42	+4.05 GRC	+0.30	+3.92 GRC
	60%, first adj. year, 10 years	−0.24	−1.04 FIN	0.00	0.00	0.00	0.00
	70%, first adj. year, 10 years	+0.33	+1.45 GRC	+0.07	+1.42 GRC	+0.06	+1.26 GRC
	80%, first adj. year, 10 years	+0.99	+3.18 GRC	+0.26	+3.16 GRC	+0.20	+3.01 GRC
Adverse $r - g$ shock	0.25 pp.	−0.27	−0.49 ITA	−0.04	−0.21 BEL	−0.01	−0.14 ESP
	0.75 pp.	+0.26	+0.48 ITA	+0.16	+0.44 GRC	+0.09	+0.36 GRC
Lower-SPB shock	0.25 pp.	−0.20	−0.28 ITA	0.00	−0.06 DEU	0.00	0.00
	0.75 pp.	+0.21	+0.28 ITA	+0.06	+0.23 PRT	+0.04	+0.21 PRT
Financial stress shock	0.5 pp.	−0.03	−0.13 ITA	0.00	0.00	0.00	0.00
	1.5 pp.	+0.03	+0.13 ITA	0.00	0.00	0.00	0.00
Ageing cost period	5 years	−0.17	−0.87 ESP	−0.19	−0.85 CZE	−0.12	−0.77 ESP
All settings	lenient	−0.17	−0.87 ESP	−0.36	−1.15 ESP	−0.16	−1.12 ESP
	strict	0.00	0.00	+0.21	+1.00 GRC	+0.12	+0.84 GRC
Data							
Repayment profile	Eurostat profile	−0.01	−0.48 GRC	−0.01	−0.36 GRC	+0.02	+0.42 FIN

Source: Author's calculations. Note: See Table 3b. With a seven-year adjustment, the debt sustainability safeguard can be met in all runs.

Table A3: Effect of selected settings of risk tolerance by country, DSA target, four-year adjustment (percentage points of GDP)

	Thr. 80% (1)	Early start (2)	Early, 80% (3)	$r - g$ 0.75 pp. (4)	Low SPB 0.75 pp. (5)	Ageing 5 yrs (6)	Lenient (7)	Strict (8)
Greece	+0.02	0.00	+1.48	+0.44	0.00	−0.41	−0.62	+0.44
Italy	0.00	0.00	+0.14	+0.49	0.00	0.00	−0.38	+0.49
France	+0.04	0.00	+0.35	+0.40	+0.02	0.00	−0.08	+0.40
Spain	0.00	0.00	0.00	+0.33	+0.05	−0.89	−1.25	+0.33
Belgium	0.00	0.00	0.00	+0.36	0.00	−0.49	−0.70	+0.36
Portugal	+0.01	0.00	+0.23	+0.27	+0.19	−0.77	−1.06	+0.27
Finland	+0.46	+0.09	+0.60	+0.10	+0.18	0.00	−0.38	+0.46
Austria	+0.11	0.00	+0.32	+0.31	+0.11	−0.22	−0.55	+0.31
Hungary	+0.01	0.00	+0.01	+0.19	+0.18	−0.50	−1.07	+0.19
Cyprus	0.00	0.00	0.00	+0.13	+0.17	−0.23	−0.42	+0.17
Slovenia	0.00	0.00	0.00	+0.12	+0.19	−0.20	−0.39	+0.19
Germany	0.00	0.00	0.00	+0.19	+0.17	−0.06	−0.26	+0.19
Croatia	0.00	0.00	0.00	0.00	0.00	+0.05	+0.05	0.00
Slovakia	0.00	0.00	0.00	+0.11	+0.19	−0.15	−0.34	+0.19
Poland	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Malta	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Romania	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Netherlands	0.00	0.00	0.00	0.00	0.00	−0.65	−0.65	0.00
Czechia	0.00	0.00	0.00	0.00	0.00	−0.66	−0.66	0.00
Latvia	0.00	0.00	0.00	0.00	0.00	+0.20	+0.20	0.00
Ireland	0.00	0.00	0.00	0.00	0.00	−0.46	−0.46	0.00
Lithuania	0.00	0.00	0.00	0.00	0.00	−0.54	−0.54	0.00
Sweden	0.00	0.00	0.00	0.00	0.00	+0.05	+0.05	0.00
Luxembourg	0.00	0.00	0.00	0.00	0.00	−0.70	−0.70	0.00
Denmark	0.00	0.00	0.00	0.00	0.00	−0.16	−0.16	0.00
Bulgaria	0.00	0.00	0.00	0.00	0.00	+0.26	+0.26	0.00
Estonia	0.00	0.00	0.00	0.00	0.00	+0.09	+0.09	0.00
Mean	+0.03	0.00	+0.15	+0.16	+0.06	−0.24	−0.40	+0.18

Source: Author's calculations. Note: Change of the DSA target relative to the baseline. (1) Probability threshold of 80 percent. (2) Stochastic projection starting in the first adjustment year. (3) Both. (4) Adverse $r - g$ shocks of 0.75 percentage points. (5) Lower-SPB shock of 0.75 percentage points. (6) Ageing cost period of five years. (7) and (8) Lenient and strict scenarios, see Table 3b. Mean: mean across the 21 countries with adjustment requirements. Countries are ordered by debt in 2024.

Table A4: Mean change of the SPB by individual check with the 2024 and the September 2026 data, four-year adjustment (percentage points of GDP)

		Own criterion		DSA target		Binding target	
Setting	Alternative	2024	2026	2024	2026	2024	2026
1. Forecasts							
Real short-term rate, $T + 10$	−1 pp.	−0.04	−0.04	−0.04	−0.04	−0.03	−0.03
	+1 pp.	+0.04	+0.04	+0.04	+0.04	+0.03	+0.04
Real long-term rate, $T + 10$	−1 pp.	−0.26	−0.26	−0.23	−0.26	−0.17	−0.20
	+1 pp.	+0.25	+0.26	+0.23	+0.25	+0.17	+0.19
Inflation, $T + 10$	−0.75 pp.	+0.18	+0.20	+0.11	+0.14	+0.10	+0.12
	+0.75 pp.	−0.16	−0.16	−0.04	−0.07	−0.03	−0.09
Potential growth, $T + 10$	−1 pp.	+0.54	+0.56	+0.42	+0.48	+0.35	+0.39
	+1 pp.	−0.51	−0.54	−0.29	−0.35	−0.20	−0.29
Ageing costs	−0.2 pp.	−0.09	−0.09	−0.08	−0.09	−0.06	−0.07
	+0.2 pp.	+0.09	+0.09	+0.08	+0.10	+0.06	+0.04
Real short-term rate, $T + 30$	−0.5 pp.	0.00	0.00	0.00	0.00	0.00	0.00
	+0.5 pp.	0.00	0.00	0.00	0.00	0.00	0.00
Real long-term rate, $T + 30$	−0.5 pp.	−0.01	−0.01	−0.01	−0.01	−0.01	0.00
	+0.5 pp.	+0.01	+0.01	+0.01	+0.01	0.00	+0.01
Inflation, $T + 30$	−0.25 pp.	+0.01	+0.01	+0.01	+0.01	0.00	+0.01
	+0.25 pp.	−0.01	−0.01	−0.01	−0.01	−0.01	−0.01
2. Model parameters							
Fiscal multiplier	0.5	−0.01	−0.01	−0.04	−0.02	−0.13	−0.16
	1	+0.01	+0.01	+0.06	+0.03	+0.22	+0.26
	1.5	+0.04	+0.04	+0.28	+0.24	+0.95	+0.82
Multiplier persistence	2 years	−0.01	−0.01	−0.03	−0.02	−0.08	−0.10
	4 years	+0.01	+0.01	+0.04	+0.02	+0.10	+0.12
	5 years	+0.03	+0.03	+0.08	+0.06	+0.18	+0.23
Budget semi-elasticity	×0.9	−0.01	−0.01	−0.01	−0.01	−0.05	−0.05
	×1.1	+0.01	+0.01	+0.01	+0.01	+0.06	+0.06
3. Estimation of risk							
Start of stochastic projection	first adjustment year	+0.20	+0.23	0.00	+0.01	0.00	0.00
Horizon of stochastic projection	10 years	+0.14	+0.14	−0.01	−0.01	0.00	0.00
Start of shock sample	1990	−0.02	−0.01	0.00	0.00	0.00	0.00
	2010	+0.01	+0.01	0.00	0.00	0.00	0.00
Shock frequency	annual	−0.14	−0.17	0.00	0.00	0.00	0.00
Shock distribution	VAR bootstrap	+0.06	+0.02	+0.02	+0.03	0.00	0.00
Winsorised shocks	P1 and P99	+0.52	+0.51	+0.04	+0.02	+0.02	+0.01
	off	+0.83	+0.81	+0.09	+0.11	+0.04	+0.06
4. Risk tolerance							
Probability threshold	60%	−0.49	−0.54	−0.01	−0.01	0.00	0.00
	80%	+0.58	+0.62	+0.03	+0.07	+0.01	+0.02
Adverse $r - g$ shock	0.25 pp.	−0.28	−0.27	−0.06	−0.06	−0.05	−0.05
	0.75 pp.	+0.28	+0.27	+0.16	+0.18	+0.12	+0.14
Financial stress shock	0.5 pp.	−0.02	−0.02	0.00	0.00	0.00	0.00
	1.5 pp.	+0.02	+0.02	0.00	0.00	0.00	0.00
Lower-SPB shock	0.25 pp.	−0.23	−0.23	0.00	−0.01	−0.01	−0.01
	0.75 pp.	+0.23	+0.23	+0.06	+0.10	+0.04	+0.07
Ageing cost period	5 years	−0.22	−0.20	−0.24	−0.23	−0.20	−0.18
Scenario	lenient	−0.22	−0.20	−0.40	−0.43	−0.27	−0.32
	strict	0.00	0.00	+0.18	+0.23	+0.13	+0.16
Data							
Repayment profile	Eurostat profile	−0.01	+0.04	−0.01	+0.04	0.00	+0.04

Source: Author's calculations. Note: Mean change across the 21 countries with adjustment requirements in the 2024 data, relative to the baseline of each data vintage. 2024: Commission prior guidance data, adjustment from 2025 to 2028. 2026: September 2026 data, adjustment from 2027 to 2030.

Table A5: SPB targets with the September 2026 data (percent of GDP)

	2026		Four-year adjustment, 2027–2030					Change vs. 2024 data			Seven-year
	Debt	SPB	DSA target	DSA criterion	Binding target	Binding constraint	Annual adj.	DSA target	Binding target	Annual adj.	Binding target
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Greece	140.7	2.54	2.05	Adverse $r-g$ test: debt declines	2.54	Adjustment floor	0.00	−0.26	−0.04	−0.21	2.54
Italy	138.5	0.97	3.18	Adverse $r-g$ test: debt declines	3.21	DSA criteria	0.56	−0.11	−0.10	−0.53	3.00
France	118.1	−1.97	1.77	Adverse $r-g$ test: debt declines	1.79	DSA criteria	0.94	+1.03	+1.04	+0.01	2.49
Belgium	110.5	−1.96	1.87	Adverse $r-g$ test: debt declines	1.88	DSA criteria	0.96	+0.70	+0.67	+0.15	2.32
Spain	99.6	−0.40	2.82	Adverse $r-g$ test: debt declines	2.84	DSA criteria	0.81	+0.25	+0.25	−0.05	2.96
Finland	91.2	−1.31	1.83	Stochastic	8.49	Debt safeguard	2.45	+0.04	+3.66	+1.11	10.87
Portugal	87.6	2.20	2.59	Adverse $r-g$ test: debt declines	2.60	DSA criteria	0.10	+0.10	+0.08	+0.02	2.34
Austria	83.4	−1.72	1.13	Adverse $r-g$ test: debt declines	1.16	DSA criteria	0.72	+0.32	−1.06	−0.17	1.66
Hungary	75.1	−2.01	3.21	Adverse $r-g$ test: debt declines	3.23	DSA criteria	1.31	+0.57	+0.56	+0.65	3.66
Germany	65.8	−1.77	1.10	Adverse $r-g$ test: debt declines	2.75	Debt safeguard	1.13	+0.11	+0.29	+0.26	2.29
Slovenia	64.9	−2.40	1.00	Lower-SPB test: debt below 60% at end	1.04	DSA criteria	0.86	+0.52	+0.52	+0.42	1.17
Poland	64.5	−3.77	1.75	Adverse $r-g$ test: debt declines	1.79	DSA criteria	1.39	+1.18	+1.21	+0.59	2.32
Slovakia	63.7	−2.63	1.67	Lower-SPB test: debt below 60% at end	1.69	DSA criteria	1.08	+0.86	+0.87	−0.19	1.85
Romania	61.6	−2.39	1.46	Lower-SPB test: debt below 60% at end	1.49	DSA criteria	0.97	+1.20	+1.22	−0.20	2.23
Croatia	55.9	−1.74	−0.95	Baseline: debt below 60% at end	−0.14	Deficit resilience	0.40	−0.49	+0.22	0.00	0.01
Cyprus	50.4	2.52	0.10	Deficit below 3%	0.12	DSA criteria	−0.60	−0.25	−3.36	−0.60	−0.35
Latvia	48.8	−2.08	0.01	Lower-SPB test: debt below 60% at end	0.04	DSA criteria	0.53	+1.29	+0.18	+0.22	0.30
Netherlands	46.9	−0.45	−0.01	Deficit below 3%	−0.01	DSA criteria	0.11	−0.06	−0.08	−0.07	−0.31
Malta	46.2	−0.29	−0.94	Deficit below 3%	0.19	Deficit resilience	0.12	+0.29	+1.06	−0.39	0.06
Czechia	45.8	−1.33	0.98	Deficit below 3%	0.99	DSA criteria	0.58	+0.61	+0.60	+0.46	1.40
Lithuania	44.6	−1.05	0.47	Deficit below 3%	0.54	Deficit resilience	0.40	+0.23	+0.27	+0.24	0.55
Sweden	36.6	−1.35	−1.55	Deficit below 3%	−0.65	Deficit resilience	0.17	+0.23	+0.09	+0.53	−0.47
Ireland	32.4	2.09	−0.46	Deficit below 3%	−0.43	DSA criteria	−0.63	+0.20	+0.20	+0.16	−0.36
Bulgaria	32.3	−2.93	−1.18	Deficit below 3%	−1.17	DSA criteria	0.44	+0.89	−0.29	+0.14	−0.76
Luxembourg	29.2	0.75	0.18	Deficit below 3%	0.19	DSA criteria	−0.14	+0.42	+0.41	−0.06	0.05
Denmark	27.0	2.75	−1.29	Deficit below 3%	−1.25	Adjustment floor	−1.00	+0.12	−0.17	0.00	−0.84
Estonia	26.9	−3.20	−1.28	Deficit below 3%	−1.28	DSA criteria	0.48	+0.63	−0.99	+0.48	−0.89

Source: Author's calculations based on the September 2026 input data. Note: Columns (1) and (2) are debt and the SPB in 2026, the reference year of this data vintage. Columns (3) to (7) refer to a four-year adjustment period ending in 2030: the DSA target and the DSA criterion that sets it, the binding target and the constraint that sets it, and the annual adjustment required by the binding target. Columns (8) to (10) are the changes of the targets and of the annual adjustment relative to the baseline with the 2024 data, whose targets refer to 2028. Column (11) is the binding target at the end of a seven-year adjustment period (2033). Countries are ordered by debt in 2026.

Table A6: Fiscal multiplier checks with the persistent multiplier and the Commission's output gap rule, binding target, four-year adjustment (percent of GDP and percentage points)

	Austria		Finland		Germany	
	Persistent	Commission	Persistent	Commission	Persistent	Commission
Baseline target	2.22	2.58	4.83	5.99	2.46	2.78
Fiscal multiplier: 0.5	−0.64	−0.80	−1.00	−1.44	−0.52	−0.64
Fiscal multiplier: 1	+1.04	+1.52	+1.64	+2.60	+0.76	+1.00
Fiscal multiplier: 1.5	+8.56	inf.	inf.	inf.	+4.08	+7.00
Multiplier persistence: 2 years	−0.40	−0.64	−0.64	−1.36	−0.32	−0.56
Multiplier persistence: 4 years	+0.44	+0.48	+0.68	+1.04	+0.32	+0.40
Multiplier persistence: 5 years	+0.76	+0.88	+1.20	+1.88	+0.56	+0.68

Source: Author's calculations. Note: The first row is the binding target of the baseline. The other rows are the change of the binding target relative to the baseline. inf.: the debt sustainability safeguard cannot be met with an annual adjustment of up to 5 percentage points. Persistent: multiplier effect on the output gap fading over the persistence period, as in the rest of this paper. Commission: output gap closure rule of the Commission's prior guidance calculation sheets. The debt sustainability safeguard sets the binding target of the three countries in the baseline.

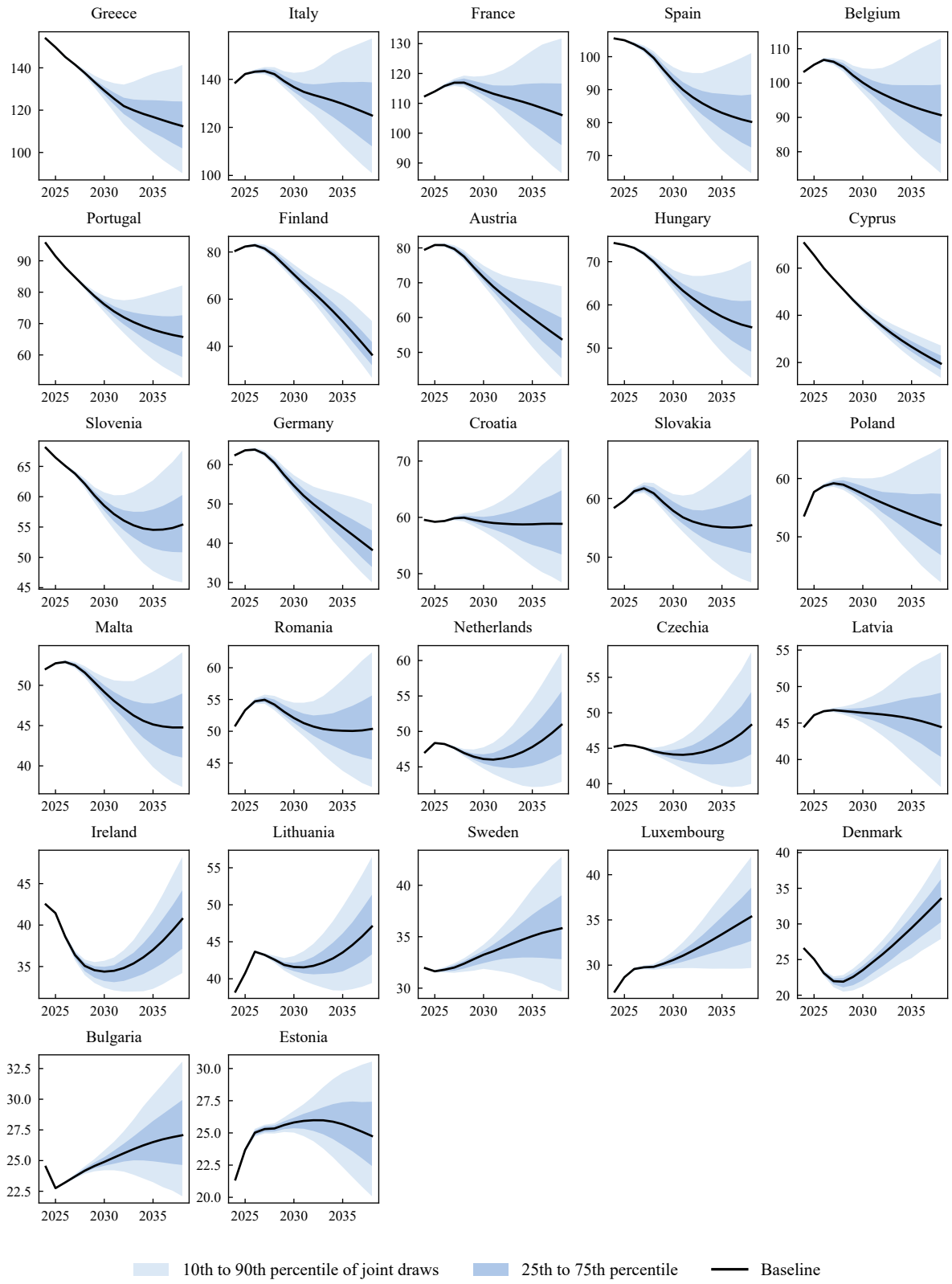


Figure A1: Debt under the baseline SPB path in the joint draws, four-year adjustment (percent of GDP)

Source: Author's calculations. Note: Debt when the SPB follows the path of the baseline binding target in all draws, with the adjustment from 2025 to 2028 including the minimum adjustment of the EDP and the deficit resilience safeguard. Only the forecasts and the model parameters differ across draws. The black line shows the baseline. The shaded areas show the range between the 10th and the 90th percentile (light) and between the 25th and the 75th percentile (dark) of debt in each year across the 500 joint draws of forecasts and model parameters. Countries are ordered by debt in 2024.